Trijet cross section at large N_c (Massive case)

Section 4 (new) Fourier transforms the dipole-only terms of the cross sections to momentum space.

1 Setup

The cross section for trijet production is given by

$$\frac{d^3N}{d^3k_1\,d^3k_2\,d^3k_3}\,(2\pi)\,\delta\left(q^+-k_1^+-k_2^+-k_3^+\right)=\frac{1}{2}\int_{w,w'}e^{-iq\cdot(w-w')}\,\langle\gamma_T^i(Q,q^+,w')|\,\mathcal{N}_q(k_1)\,\mathcal{N}_{\bar{q}}(k_2)\,\mathcal{N}_g(k_3)\,|\gamma_T^i(Q,q^+,w)\rangle_{q\bar{q}g}\quad(1.1)$$

1.1 Kinematics

The virtual photon with the four momentum as $q^\mu=\left(q^+,-\frac{Q^2}{2q^+},q\right)$. Let us define the kinematics in the following: The onshell

Component	Longitudinal momentum fraction	Transverse momentum	mass
Quark	$\zeta=\frac{k_1^+}{q^+}$	k_1	m_f
Anti Quark	$\eta=1-\zeta-\xi$	k_2	m_f
Gluon	$\xi=\frac{k_3^+}{q^+}$	k_3	0

denominators are $E_q(k)=\frac{k^2+m_f^2}{2k^+}$, $E_{\bar{q}}(p)=\frac{p^2+m_f^2}{2p^+}$, $E_g(p)=\frac{p^2}{2p^+}$.

1.2 Matrix Elements

The matrix elements that are used in this process are as follows:

1. Virtual photon splitting into a quark and anti quark pair

$$\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)|\,H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle=(2\pi)^3\,\delta^{(3)}(q-k-p)\,\frac{e\,e_f\,\delta_{\alpha\beta}}{2\,\sqrt{2q^+}}\,\chi^\dagger_{\lambda_1}\,\left[\frac{2q^\lambda}{q^+}-\sigma^\lambda\,\frac{\sigma\cdot p+m_f\,\sigma^3}{p^+}-\frac{\sigma\cdot k-m_f\,\sigma^3}{k^+}\sigma^\lambda\right]\,\chi_{\lambda_2}\quad(1.2)$$

rewriting it using the delta function,

$$k^+=\zeta\,q^+,\,p^+=(1-\zeta)\,q^+,\,\tilde{k}=k-\zeta\,q$$

we get

$$\left[\frac{2q^\lambda}{q^+}-\frac{\sigma\cdot k-m_f\,\sigma^3}{k^+}\,\sigma^\lambda-\sigma^\lambda\,\frac{\sigma\cdot(q-k)+m_f\,\sigma^3}{(q^+-k^+)}\right]=\frac{\left[(2\zeta-1)\,\delta^{\lambda\,j}+i\epsilon^{\lambda\,j}\sigma^3\right]\,\tilde{k}^j+m_f\,\sigma^3\,\sigma^\lambda}{\zeta\,(1-\zeta)\,q^+}$$

where we used $\sigma^i\sigma^j=\delta^{ij}+i\epsilon^{ij}\sigma^3$ and $\sigma^\lambda\sigma^3=-\sigma^3\sigma^\lambda$ and

$$\chi^\dagger_{\lambda_1}\left[(2\zeta-1)\,\delta^{\lambda\,j}+i\epsilon^{\lambda\,j}\sigma^3\right]\chi_{\lambda_2}=\delta_{\lambda_1\,\lambda_2}\left[(2\zeta-1)\delta^{\lambda\,j}+2i\epsilon^{\lambda\,j}\,\lambda_1\right]$$

$$\chi^\dagger_{\lambda_1}\,\sigma^3\sigma^\lambda\,\chi_{\lambda_2}=\delta_{\lambda_2,-\lambda_1}\left[2\lambda_1\delta^{\lambda,1}-i\delta^{\lambda,2}\right]\equiv\mu^\lambda_{\lambda_1\lambda_2}$$

The mass term flips the helicity ($\lambda_2=-\lambda_1$). The matrix element defined above can be rewritten as follows:

$$\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)|\,H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle=(2\pi)^3\,\delta^{(3)}(q-k-p)\,\frac{e\,e_f\,\delta_{\alpha\beta}\left\{\left[(2\zeta-1)\,\delta^{\lambda\,j}+2i\epsilon^{\lambda\,j}\,\lambda_1\right]\,\delta_{\lambda_1\,\lambda_2}\tilde{k}^j+m_f\,\mu^\lambda_{\lambda_1\lambda_2}\right\}}{2\,\sqrt{2\,q^{+\,3}}\,\zeta\,(1-\zeta)}\quad(1.3)$$

2. Gluon emission from anti - quark

$$\langle q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\,g^{a'}_{i'}(q')|\,H_{\bar{q}g}\,|q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)\rangle=-\,(2\pi)^6\,\delta^{(3)}(k-k')\,\delta^{(3)}(p-p'-q')\,\delta_{\alpha'\,\alpha}\,\delta_{\lambda_1\,\lambda_1'}\,\frac{g\,t^{a'}_{\beta\,\beta'}}{2\sqrt{2\,q'^+}}$$
$$\chi^\dagger_{\lambda_2}\,\left[\frac{2\,q'^{i'}}{q'^+}-\sigma^{i'}\,\frac{\sigma\cdot p'+m_f\,\sigma^3}{p'^+}-\frac{\sigma\cdot p+m_f\,\sigma^3}{p^+}\,\sigma^{i'}\right]\,\chi_{\lambda_2'}\quad(1.4)$$

with $p^+ = (1-\zeta)\,q^+$, $p'^+ = \eta\,q^+$, $q'^+ = \xi\,q^+$ and $\tilde{p} = q' - \frac{\xi}{(1-\zeta)}\,p$. The terms proportional to p cancel and from here onwards we term $\omega_f = \frac{\xi\,m_f}{(1-\zeta)}$. The reduced vertex is

$$\chi^\dagger_\lambda \left[\frac{2\,q'^{i'}}{q'^+} - \sigma^{i'}\,\frac{\sigma\cdot p' + m_f\,\sigma^3}{p'^+} - \frac{\sigma\cdot p + m_f\,\sigma^3}{p^+}\,\sigma^{i'}\right]\chi_{\lambda_2} = \frac{\tau^{i'n}_{\lambda\lambda_2}(\xi,\eta)\,\tilde{p}^n + \xi\,\omega_f\,\mu^{i'}_{\lambda\lambda_2}}{\xi\,\eta\,q^+} \tag{1.5}$$

and

$$m_f\,\sigma^3\,\sigma^{i'}\left[\frac{1}{p'^+} - \frac{1}{p^+}\right] = \frac{\xi^2\,m_f}{(1-\zeta)}\,\frac{\sigma^3\sigma^{i'}}{\xi\,\eta\,q^+} \tag{1.6}$$

3. Instantaneous contribution

$$\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(s)\,g^b_j(p)|\;H_{\gamma\to q\bar{q}g}\,|\gamma\rangle = -(2\pi)^3\,\delta^{(3)}(q-k-p-s)\,\frac{g\,e\,e_f\,\delta_{\lambda_1\lambda_2}\,t^b_{\alpha\beta}}{4\,\sqrt{q^+p^+}}\,\frac{\delta^{ij} + 2i\,\epsilon^{ij}\lambda_1}{p^+ + s^+} \tag{1.7}$$

1.3 Notations

$$\tilde{k} = k - \zeta\,q, \qquad \tilde{p} = p - \frac{\xi}{1-\zeta}\,(q-k), \qquad \tilde{Q}^2 = \zeta(1-\zeta)\,Q^2$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta\,(1-\zeta)\,Q^2 + m_f^2, \qquad \omega_f = \frac{\xi\,m_f}{(1-\zeta)}, \qquad E_f^2 = Q^2 + m_f^2\left(\frac{1}{\zeta} + \frac{1}{\eta}\right), \qquad a = \frac{\xi\,\eta}{\zeta\,(1-\zeta)^2}$$

$$\varphi^{ij}_{\lambda_1\lambda}(\zeta) = \delta_{\lambda_1\lambda}\left[(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1\right], \qquad \tau^{mn}_{\lambda\lambda_2}(\xi,\eta) = \delta_{\lambda\lambda_2}\,\left[(2(1-\zeta)-\xi)\delta^{mn} + 2i\,\xi\,\epsilon^{mn}\,\lambda\right]$$

$$\mu^i_{\lambda\lambda'} = \delta_{\lambda',-\lambda}\left[2\lambda\delta^{i1} - i\delta^{i2}\right] = \chi^\dagger_\lambda\sigma^3\sigma^i\chi_{\lambda'}, \qquad \sum_i \mu^i_{\lambda\lambda'}\mu^{i*}_{\lambda\lambda'} = 2\delta_{\lambda',-\lambda}$$

Here ϵ_f is the usual mass shifted photon scale. The photon vertex and the anti quark - gluon vertex are:

$$\text{photon}: \varphi^{ij}_{\lambda_1\lambda}(\zeta)\tilde{k}^j + m_f\mu^i_{\lambda_1\lambda}, \qquad \text{gluon off the antiquark}: \tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,\tilde{p}^n + \xi\,\omega_f\,\mu^m_{\lambda\lambda_2}$$

The energy denominator is as follows:

$$E_q + E_{\bar{q}} - E_\gamma = \frac{\tilde{k}^2 + \tilde{Q}^2 + m_f^2}{2\zeta\,(1-\zeta)\,q^+} = \frac{\tilde{k}^2 + \epsilon_f^2}{2\,\zeta(1-\zeta)\,q^+}$$

$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_\gamma = \frac{(1-\zeta)(\tilde{p}^2 + \omega_f^2)}{2\,\xi\,\eta\,q^+} + \frac{(\tilde{k}^2 + \epsilon_f^2)}{2\zeta\,(1-\zeta)\,q^+} = \frac{\zeta\,(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\,\eta\,(\tilde{k}^2 + \epsilon_f^2)}{2\,\zeta\,(1-\zeta)\,\xi\,\eta\,q^+}$$

$$\begin{aligned} E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} &= \frac{p^2}{2q^+\,\xi} + \frac{(q-k-p)^2 + m_f^2}{2q^+\,\eta} - \frac{(q-k)^2 + m_f^2}{2q^+\,(1-\zeta)} = \frac{\left[(1-\zeta)\,p - \xi\,(q-k)\right]^2}{2\xi\,(1-\zeta)\,\eta\,q^+} + \frac{m_f^2}{2q^+}\left(\frac{1}{\eta} - \frac{1}{1-\zeta}\right) \\ &= \frac{(1-\zeta)^2\,\tilde{p}^2 + \xi^2\,m_f^2}{2\,\xi(1-\zeta)\,\eta\,q^+} \end{aligned} \tag{1.8}$$

Simplifying it we get

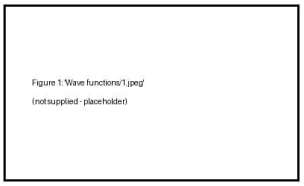
$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} = \frac{(1-\zeta)\,(\tilde{p}^2 + \omega_f^2)}{2\xi\,\eta\,q^+}$$

Also, for instantaneous term involving gluon emission is given by

$$\zeta(1-\zeta)^2\,\omega_f^2 + \xi\,\eta\,\epsilon_f^2 = \zeta\,(1-\zeta)\,\xi\,\eta\,E_f^2, \qquad E_f^2 = Q^2 + \frac{m_f^2}{\zeta} + \frac{m_f^2}{\eta} \tag{1.9}$$

2 Outgoing component of wave function through LCPT

1. Anti quark emits a gluon before the shock wave



$$|\gamma\rangle^{(1)}_{q\bar{q}g} = |q_2\bar{q}_2g_2\rangle\;\langle q_2\bar{q}_2g_2|\;S\;|q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1|\,H_{\bar{q}g}\,|q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q}|\,H_{\gamma\to q\bar{q}}\,|\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \tag{2.1}$$

$$|\gamma\rangle^{(1)}_{q\bar{q}g} = \int d^3k''\, d^3p''\, d^3r\, d^3k'\, d^3p'\, d^3q'\, d^3k\, d^3p\; |q^{\alpha''}_{\lambda_1''}(k'')\, \bar{q}^{\beta''}_{\lambda_2''}(p'')\, g^b_j(r)\rangle\; \langle q^{\alpha''}_{\lambda_1''}(k'')\, \bar{q}^{\beta''}_{\lambda_2''}(p'')\, g^b_j(r)|\; S\; |q^{\alpha'}_{\lambda_1'}(k')\, \bar{q}^{\beta'}_{\lambda_2'}(p')\, g^{a'}_{i'}(q')\rangle \times$$

$$\frac{\langle q^{\alpha'}_{\lambda_1'}(k')\, \bar{q}^{\beta'}_{\lambda_2'}(p')\, g^{a'}_{i'}(q')|\, H_{\bar{q}g}\, |q^{\alpha}_{\lambda_1}(k)\, \bar{q}^{\beta}_{\lambda_2}(p)\rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q^{\alpha}_{\lambda_1}(k)\, \bar{q}^{\beta}_{\lambda_2}(p)|\, H_{\gamma\to q\bar{q}}\, |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q,Q)} \tag{2.2}$$

We will first evaluate the matrix elements without scattering (i.e., $S = 1$). With the massive vertices and energy denominators of Sec. 1, we get the following

$$|\gamma_i\,(Q,q^+,q)\rangle^{(1\,a)}_{q\bar{q}g} = -\frac{1}{(2\pi)^6}\int d^2\tilde{k}\, d^2\tilde{p}\int_0^1 d\zeta \int_0^{1-\zeta} d\xi\; |q^{\alpha}_{\lambda_1}(\zeta q^+,k)\bar{q}^{\beta}_{\lambda_2}(\eta\, q^+, q-k-p) g^a_m(\xi q^+,p)\rangle\, \frac{e\, e_f\, g\, q^+\, \zeta(1-\zeta)}{2\sqrt{\xi}\,(\tilde{k}^2+\epsilon_f^2)}$$

$$\frac{\left[\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\tilde{k}^j + m_f\,\mu^i_{\lambda_1\lambda}\right]\,\left[\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,\tilde{p}^n\, + \xi\,\omega_f\,\mu^m_{\lambda\lambda_2}\right]\,t^a_{\alpha\beta}}{\left[\zeta(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2) + \xi\,\eta\,(\tilde{k}^2+\epsilon_f^2)\right]} \tag{2.3}$$

However, Wilson lines can be obtained only if the wave function is expressed in transverse coordinates. Therefore, we perform a Fourier transform to rewrite the wave function in transverse coordinate space. Performing Fourier transform for transverse co-ordinates,

$$|q^{\alpha}_{\lambda_1}(k)\bar{q}^{\beta}_{\lambda_2}(q-k-p)g^a_m(p)\rangle = \int_{x,y,z} e^{ik\cdot x}\, e^{ip\cdot z}\, e^{i(q-k-p)\cdot y}\; |q^{\alpha}_{\lambda_1}(x)\bar{q}^{\beta}_{\lambda_2}(y)g^a_m(z)\rangle \tag{2.4}$$

Rewriting k, p in terms of $\tilde{k}$ and $\tilde{p}$ respectively, we get

$$\begin{aligned} e^{i[\zeta\, x+\eta\, y+\xi\, z]\cdot q}\, e^{-i\,(y-z)\cdot\tilde{p}}\, e^{i[(1-\zeta)x-\eta y-\xi z]\cdot\frac{\tilde{k}}{(1-\zeta)}} &= e^{i\,(\zeta\, x+\eta\, y+\xi\, z)\cdot q}\, e^{-i\,(y-z)\cdot\tilde{p}}\, e^{i\left(x-\frac{\eta\, y+\xi\, z}{(1-\zeta)}\right)\cdot\tilde{k}} \\ &= e^{i\,w\cdot q}\, e^{-i\,Y\cdot\tilde{p}}\, e^{i\,R\cdot\tilde{k}}. \end{aligned} \tag{2.5}$$

The transverse coordinates of the photon (w) and the intermediate antiquark (y', before emitting the gluon) are fixed by centre of energy conservation at the two vertices:

$$w = \zeta\, x + (1-\zeta)\, y' = \zeta\, x + \eta\, y + \xi\, z, \tag{2.6}$$

and

$$y' = \frac{\eta\, y + \xi\, z}{(1-\zeta)}. \tag{2.7}$$

The relative distances are defined as

$$R = x - y'; \qquad Y = y - z$$

where we used

$$(1-\zeta)\, x - \eta\, y - \xi\, z = (1-\zeta)\,(x-y') = (1-\zeta)\, R$$

We will use the following Fourier transforms:

$$\int \frac{d^2k}{2\pi}\,\frac{d^2p}{2\pi}\,\frac{k^i p^j}{k^2\,[ak^2+bp^2]}\, e^{ik\cdot X + ip\cdot Z} = -\frac{X^i Z^j}{Z^2\,[bX^2+aZ^2]} \tag{2.8}$$

With masses the two denominators contain two different scales ϵ_f and ω_f and the transform is no longer a single Bessel function. We need four transforms, one with each helicity path. Define

$$(1-\zeta)\int \frac{d^2\tilde{k}}{2\pi}\,\frac{d^2\tilde{p}}{2\pi}\,\frac{e^{i\tilde{k}\cdot R}\, e^{-i\tilde{p}\cdot Y}\,\{\tilde{k}^j\tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1\}}{(\tilde{k}^2+\epsilon_f^2)\,\left[\zeta\,(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2) + \xi\,\eta\,(\tilde{k}^2+\epsilon_f^2)\right]} = \left\{R^j Y^n \mathcal{G}^{(11)}, iR^j\mathcal{G}^{(10)}, -iY^n\,\mathcal{G}^{(01)}, \mathcal{G}^{(00)}\right\} \tag{2.9}$$

Using Schwinger parametrization, we get

$$\frac{1}{(\tilde{k}^2+\epsilon_f^2)\,\left[\zeta\,(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2) + \xi\,\eta\,(\tilde{k}^2+\epsilon_f^2)\right]} = \frac{1}{\xi\,\eta}\int_0^\infty du\int_0^\infty dv\, e^{-(u+v)\,(\tilde{k}^2+\epsilon_f^2) - \frac{v}{a}(\tilde{p}^2+\omega_f^2)} \tag{2.10}$$

where $a = \frac{\xi\,\eta}{[\zeta\,(1-\zeta)^2]}$, K_ν is the Modified Bessel function of the second kind. Using the Gaussian integrals and including massive terms, we define the following notations, with x as a Feynman parameter.

$$\mathcal{G}^{(\alpha\beta)}(R,Y) = \frac{a^\beta}{2\,\zeta(1-\zeta)}\int_0^1 \frac{dx}{x^{1+\beta}}\left(\frac{A_x}{\Delta_x}\right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{A_x\Delta_x}), \alpha,\beta\in\{0,1\}$$

$$A_x = \epsilon_f^2 + \frac{x\,\omega_f^2}{a} = \zeta\,(1-\zeta)\,Q^2 + m_f^2\,\left(1 + \frac{x\zeta\xi}{\eta}\right)$$

$$\Delta_x = R^2 + \frac{a\,Y^2}{x}$$

Explicitly

$$\mathcal{G}^{(11)} = \frac{a}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x^2}\,\frac{A_x}{\Delta_x}\,K_2\left(\sqrt{A_x\Delta_x}\right) \tag{2.11}$$

$$\mathcal{G}^{(10)} = \frac{1}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x}\,\sqrt{\frac{A_x}{\Delta_x}}\,K_1\left(\sqrt{A_x\Delta_x}\right) \tag{2.12}$$

$$\mathcal{G}^{(01)} = \frac{a}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x^2}\,\sqrt{\frac{A_x}{\Delta_x}}\,K_1\left(\sqrt{A_x\,\Delta_x}\right) \tag{2.13}$$

$$\mathcal{G}^{(00)} = \frac{1}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x}\,K_0\left(\sqrt{A_x\Delta_x}\right) \tag{2.14}$$

Now, implementing all these steps in the outgoing component of wave function, we get

$$|\gamma_i\,(Q,q^+,q)\rangle^{(1\,a)}_{q\bar{q}g} = -\,\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\;\langle q^\alpha_{\lambda_1}(\zeta q^+,x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y)g^a_m(\xi q^+,z)\rangle\;\frac{\zeta\,t^a_{\alpha\beta}}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\;\times\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}] \tag{2.15}$$

where

$$\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}] = \sum_\lambda\Big[\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,R^j\,Y^n\,\mathcal{G}^{(11)} + i\,\xi\,\omega_f\,\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\mu^m_{\lambda\lambda_2}\,R^j\,\mathcal{G}^{(10)}$$
$$-\,i\,m_f\,\mu^i_{\lambda_1\lambda}\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,Y^n\,\mathcal{G}^{(01)} + m_f\,\xi\,\omega_f\,\mu^i_{\lambda_1\lambda}\,\mu^m_{\lambda\lambda_2}\,\mathcal{G}^{(00)}\Big] \tag{2.16}$$

$$\mathcal{D}^2(x,y,z,\zeta,\xi) = \zeta\,(1-\zeta)\,R^2 + \frac{\xi\,\eta\,Y^2}{(1-\zeta)} = \zeta\,\eta\,(x-y)^2 + \zeta\,\xi\,(x-z)^2 + \eta\,\xi\,(y-z)^2$$

$$\mu^i_{\lambda\lambda'} = \delta_{\lambda',-\lambda}\left[2\lambda\delta^{i1} - i\delta^{i2}\right] = \chi^\dagger_\lambda\sigma^3\sigma^i\chi_{\lambda'}$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta\,(1-\zeta)\,Q^2 + m_f^2, \qquad \omega_f = \frac{\xi\,m_f}{(1-\zeta)}, \qquad E_f^2 = Q^2 + m_f^2\left(\frac{1}{\zeta}+\frac{1}{\eta}\right), \qquad a = \frac{\xi\,\eta}{\zeta\,(1-\zeta)^2}$$

$$\varphi^{ij}_{\lambda_1\lambda}(\zeta) = \delta_{\lambda_1\lambda}\left[(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1\right], \qquad \tau^{mn}_{\lambda\lambda_2}(\xi,\eta) = \delta_{\lambda\lambda_2}\;[(2(1-\zeta)-\xi)\delta^{mn} + 2i\,\xi\,\epsilon^{mn}\,\lambda]$$

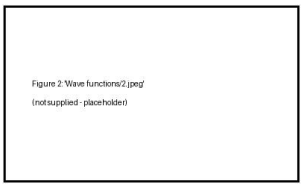
and

$$c = \zeta\,x + \eta\,y + \xi\,z$$

is the centre of energy of trijet system. Attaching Wilson lines, we get

$$|\gamma_i\,(Q,q^+,q)\rangle^{(1\,a)}_{q\bar{q}g} = -\,\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\;\langle q^\alpha_{\lambda_1}(\zeta q^+,x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y)g^a_m(\xi q^+,z)\rangle\;\frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\;\times\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}](R,Y)$$
$$\left[U^{ab}(z)\,V(x)\,t^b\,V^\dagger(y) - t^a\right]_{\alpha\beta} \tag{2.17}$$

2. Anti quark emits a gluon after the shock wave



$$|\gamma\rangle^{(2)}_{q\bar{q}g} = |q_2\bar{q}_2g_2\rangle\;\langle q_2\bar{q}_2g_2|\;S\;|q_1\bar{q}_1g_1\rangle\times\frac{\langle q_1\bar{q}_1g_1|\,H_{\bar{q}g}\,|q\bar{q}\rangle}{E_{\bar{q}_1}+E_{q_1}-E_{q_2}-E_{\bar{q}_2}-E_{g_2}}\times\frac{\langle q\bar{q}|\,H_{\gamma\to q\bar{q}}\,|\gamma\rangle}{E_q+E_{\bar{q}}-E_\gamma} \tag{2.18}$$

$$|\gamma\rangle^{(2)}_{q\bar{q}g} = \int d^3k''\,d^3p''\,d^3r\,d^3k'\,d^3p'\,d^3k\,d^3p\;|q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)\rangle\,\frac{\langle q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)|\,H_{\bar{q}g}\,|q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\rangle}{E_q(k')+E_{\bar{q}}(p')+E_q(k'')-E_{\bar{q}}(p'')-E_g(r)}\times$$
$$\langle q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')|\,S\,|q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)\rangle\times\frac{\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)|\,H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p)+E_q(k)-E_\gamma(q,Q)} \tag{2.19}$$

The only difference between the first and the second wave function component is the energy denominator and the Wilson line structure. The changes are as follows:

$$\frac{1}{E_{q\bar{q}g} - E_\gamma} \to \frac{1}{E_{q\bar{q}} - E_{q\bar{q}g}}$$

Equivalently,

$$\frac{1}{\zeta\,(1-\zeta)^2(\tilde{p}^2+\omega_f^2) + \xi\,\eta\,(\tilde{k}^2+\epsilon_f^2)} \to -\frac{1}{\zeta\,(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2)}$$

Also, the Fourier transform which we will be using here is the following:

$$\int \frac{d^2\tilde{k}}{2\pi}\, e^{i\tilde{k}\cdot R}\, \frac{\{\tilde{k}^j, 1\}}{\tilde{k}^2+\epsilon_f^2} = \left\{ \frac{i\,R^j}{R}\,\epsilon_f\,K_1(\epsilon_f\,R), K_0(\epsilon_f R) \right\}$$

and

$$\int \frac{d^2\tilde{p}}{2\pi}\, e^{-i\tilde{p}\cdot Y}\, \frac{\{\tilde{p}^n, 1\}}{(\tilde{p}^2+\omega_f^2)} = \left\{ -\frac{i\,Y^n}{Y}\,\omega_f\,K_1(\omega_f\,Y), K_0(\omega_f\,Y) \right\}$$

With the same normalization, we get

$$(1-\zeta)\int \frac{d^2\tilde{k}}{2\pi}\,\frac{d^2\tilde{p}}{2\pi}\, e^{i\tilde{k}\cdot R}\, e^{-i\tilde{p}\cdot Y}\, \frac{\left\{\tilde{k}^j\tilde{p}^n,\, \tilde{k}^j, \tilde{p}^n,\, 1\right\}}{(\tilde{k}^2+\epsilon_f^2)\,\zeta(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2)} = \left\{ R^j Y^n \mathcal{A}^{(11)},\, i\,R^j\,\mathcal{A}^{(10)},\, -iY^n\,\mathcal{A}^{(01)},\, \mathcal{A}^{(00)} \right\} \tag{2.20}$$

where:

$$\mathcal{A}^{(11)} = \frac{\epsilon_f\,K_1(\epsilon_f R)}{\zeta(1-\zeta)\,R}\,\frac{\omega_f\,K_1(\omega_f\,Y)}{Y}$$

$$\mathcal{A}^{(10)} = \frac{\epsilon_f\,K_1(\epsilon_f R)}{\zeta(1-\zeta)\,R}\,K_0(\omega_f Y)$$

$$\mathcal{A}^{(01)} = \frac{K_0(\epsilon_f R)}{\zeta\,(1-\zeta)}\,\frac{\omega_f\,K_1(\omega_f Y)}{Y}$$

$$\mathcal{A}^{(00)} = \frac{K_0(\epsilon_f R)}{\zeta\,(1-\zeta)}\,K_0(\omega_f Y)$$

The Wilson line will not have any component of gluon because the gluons are not getting scattered as the gluon is emitted after the scattering.

$$\left[U^{ab}(z)\,V(x)\,t^b\,V^\dagger(y) - t^a\right]_{\alpha\beta} \to \left[V(x)\,V^\dagger(y')t^a - t^a\right]_{\alpha\beta}$$

Substituting all these changes in eq. (2.17), we get

$$|\gamma(w)\rangle^{(2)}_{q\bar{q}g} = +\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\;|q^\alpha_{\lambda_1}(\zeta q^+, x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+, y)g^a_m(\xi q^+, z)\rangle\;\frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\times$$
$$\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}](R,Y)\times\left[V(x)\,V^\dagger(y')\,t^a - t^a\right]_{\alpha\beta} \tag{2.21}$$

where

$$\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}] = \sum_\lambda \Big[\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,R^j\,Y^n\,\mathcal{A}^{(11)} + i\,\xi\,\omega_f\,\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\mu^m_{\lambda\lambda_2}\,R^j\,\mathcal{A}^{(10)}$$
$$- i\,m_f\,\mu^i_{\lambda_1\lambda}\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,Y^n\,\mathcal{A}^{(01)} + m_f\,\xi\,\omega_f\,\mu^i_{\lambda_1\lambda}\,\mu^m_{\lambda\lambda_2}\,\mathcal{A}^{(00)}\Big] \tag{2.22}$$

In the massless case Eq. (2.21) follows from Eq. (2.17) by the formal substitution $(b) = (a)|_{y,z\to y'}$ of the paper (keeping the prefactor $1/Y^2$). With masses this is no longer literally true: the functions $\mathcal{A}^{(\alpha\beta)}$ are different functions from $\mathcal{G}^{(\alpha\beta)}$ (after the shock wave the gluon is emitted from an on-shell antiquark and sees only the dead-cone scale ω_f). We keep the label $[y, z \to y']$ for this term, meaning: Wilson lines at y' and $\mathcal{G} \to \mathcal{A}$. The short-distance behaviour is the same, $Y^2\mathcal{G}^{(11)}$, $Y^2\mathcal{A}^{(11)} \xrightarrow{Y\to 0} \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta)R}$, $Y^2\mathcal{G}^{(01)}$, $Y^2\mathcal{A}^{(01)} \xrightarrow{Y\to 0} \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)}$, and $\mathcal{G}^{(10)} - \mathcal{A}^{(10)}$, $\mathcal{G}^{(00)} - \mathcal{A}^{(00)}$ stay finite, so the UV cancellation between Eq. (2.17) and Eq. (2.21) works as in the massless case.

3. Instantaneous contribution (Anti quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle^{(1\,\text{inst.})}_{q\bar{q}g} = |q\bar{q}g\rangle\ \langle q\bar{q}g|\ S\ |q\bar{q}g\rangle\ \frac{\langle q\bar{q}g|\ H_{\gamma\to q\bar{q}g}\,|\gamma\rangle}{E_{q\bar{q}g}-E_\gamma} \tag{2.23}$$

Using the relevant matrix we get (In momentum space - without Wilson lines)

$$|\gamma\rangle^{(1\,\text{inst.})}_{q\bar{q}g} = \frac{1}{(2\pi)^6}\int d^2\tilde{k}\,d^2\tilde{p}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ |\bar{q}^{\beta}_{\lambda_2}(\eta\,q^+,q-k-p),g^a_m(\xi q^+,p),q^{\alpha}_{\lambda_1}(\zeta q^+,k)\rangle\ \frac{e\,e_f\,g\,q^+\,\zeta}{2\sqrt{\xi}}$$
$$\frac{(\delta^{im}+2i\,\epsilon^{im}\,\lambda_1)\,\delta_{\lambda_1\lambda_2}\,t^a_{\alpha\beta}}{\tilde{k}^2+\epsilon_f^2+\frac{\zeta\,(1-\zeta)^2}{\xi\,\eta}\,(\tilde{p}^2+\omega_f^2)} \tag{2.24}$$

Performing Fourier transform and inserting $S-1$ component, we get the following

$$(1-\zeta)\int\frac{d^2\tilde{k}}{2\pi}\,\frac{d^2\tilde{p}}{2\pi}\,e^{i\,\tilde{k}\cdot R}\,e^{-i\tilde{p}\cdot Y}\,\frac{1}{\zeta\,(1-\zeta)^2\,(\tilde{p}^2+\omega_f^2)+\xi\,\eta\,(\tilde{k}^2+\epsilon_f^2)} = \frac{E_f\,K_1(E_f\mathcal{D})}{\mathcal{D}} = \mathcal{G}_E \tag{2.25}$$

The outgoing component of wave function is obtained as follows:

$$|\gamma\rangle^{(1\,\text{inst.})}_{q\bar{q}g} = +\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ |\bar{q}^{\beta}_{\lambda_2}(\eta\,q^+,y),g^a_m(\xi\,q^+,z),q^{\alpha}_{\lambda_1}(\zeta\,q^+,x)\rangle\ \frac{\zeta\,\eta\,\sqrt{\xi}}{(1-\zeta)}\,\delta^{(2)}(w-c)\,\left(\delta^{im}+2i\,\epsilon^{im}\lambda_1\right)\,\delta_{\lambda_1\lambda_2}$$
$$\frac{E_f\,K_1(E_f\,\mathcal{D})}{\mathcal{D}}\left[U^{ab}(z)V(x)t^bV^\dagger(y)-t^a\right]_{\alpha\beta} \tag{2.26}$$

This has the same Wilson line structure as the regular i.e., Eq. (2.17). There is no instantaneous term after the shock wave (the gluon emission after the shock wave is a single vertex), so the instantaneous term has no $[y,z\to y']$ counterpart (in the massless note this was expressed by the factor $\frac{Y^2}{R\cdot Y}$ of the effective vertex, which vanishes for $y,z\to y'$). Now, we have all the wave functions corresponding to anti quark emission. All four anti quark emission can be written as a single expression with an effective vertex. The effective vertex is written as

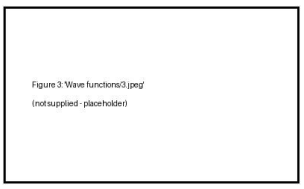
$$\mathcal{M}^{im}_{1\,\lambda_1\lambda_2}(x,y,z,\zeta,\xi) = \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}](R,Y) - \delta_{\lambda_1\lambda_2}\,\frac{\xi\,\eta}{(1-\zeta)}\,\left(\delta^{im}+2i\epsilon^{im}\,\lambda_1\right)\,\mathcal{G}_E \tag{2.27}$$

$$\mathcal{M}^{im}_{2\,\lambda_1\lambda_2}(x,y,z,\zeta,\xi) = \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}](R,Y) \tag{2.28}$$

All four anti quark emission are then written using a single expression:

$$|\gamma^i(w)\rangle_{q\bar{q}g} = -\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ |\bar{q}^{\beta}_{\lambda_2}(\eta\,q^+,y),g^a_m(\xi\,q^+,z),q^{\alpha}_{\lambda_1}(\zeta\,q^+,x)\rangle\ \frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)$$
$$\left[\mathcal{M}^{im}_{1\lambda_1\lambda_2}(x,y,z,\zeta,\xi)\left[U^{ab}(z)V(x)t^bV^\dagger(y)-t^a\right]_{\alpha\beta}-\mathcal{M}^{im}_{2\lambda_1\lambda_2}\,\left[V(x)\,V^\dagger(y')t^a-t^a\right]_{\alpha\beta}\right] \tag{2.29}$$

4. Quark emits a gluon before the shock wave



When we move from anti quark emission to quark emission, we will have to perform the following changes from anti quark wave functions:

- $\zeta\leftrightarrow\eta$
- $x\leftrightarrow y$
- $\alpha\leftrightarrow\beta$
- $\lambda_1\leftrightarrow\lambda_2$
- Change overall sign
- $y'\to x' = \frac{\zeta\,x+\xi\,z}{\zeta+\xi}$
- $R\to y-x' = -\tilde{R}$
- $Y\to x-z = X$
- $\varphi^{ij}_{\lambda_1\lambda}(\zeta)\to\varphi^{ij\,*}_{\lambda_2\lambda}(\eta) = -\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)$
- $\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\to\tau^{mn\,*}_{\lambda\,\lambda_1}(\xi,\zeta) = \tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)$
- $D(x,y,z,\zeta,\xi)\to D(y,x,z,\eta,\xi)$

- after the shock wave (massless limit): $\mathcal{D} \to \sqrt{\eta(1-\eta)}\,\tilde{R}$, i.e. $\tilde{Q} \to \sqrt{\eta(1-\eta)}\,Q$
- $m_f\,\mu^i_{\lambda_1\lambda} \to m_f\,\mu^i_{\lambda\lambda_2}$
- $\xi\,\omega_f\,\mu^m_{\lambda\lambda_2} \to \xi\,\tilde{\omega}_f\,\mu^m_{\lambda_1\lambda}$ with $\tilde{\omega}_f = \frac{\xi\,m_f}{(1-\eta)}$
- $\epsilon_f^2 \to \tilde{\epsilon}_f^2 = \eta(1-\eta)\,Q^2 + m_f^2$, $a \to \tilde{a} = \frac{\xi\,\zeta}{\eta\,(1-\eta)^2}$
- $A_x \to \tilde{A}_x = \eta\,(1-\eta)\,Q^2 + m_f^2\left(1 + \frac{x\eta\xi}{\zeta}\right)$ and $\Delta_x \to \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a}\,X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R,Y) \to \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R},X)$ and $\mathcal{A}^{(\alpha,\beta)}(R,Y) \to \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R},X)$ also, $\zeta\,(1-\zeta) \to \eta\,(1-\eta)$

$$|\gamma\rangle^{(3)}_{q\bar{q}g} = |q_2\bar{q}_2g_2\rangle\ \langle q_2\bar{q}_2g_2|\ S\ |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1|\,H_{qg}\,|q\bar{q}\rangle}{E_{\bar{q}_1}+E_{q_1}+E_{g_1}-E_\gamma} \times \frac{\langle q\bar{q}|\,H_{\gamma\to q\bar{q}}\,|\gamma\rangle}{E_q+E_{\bar{q}}-E_\gamma} \tag{2.30}$$

$$|\gamma\rangle^{(3)}_{q\bar{q}g} = \int d^3k''\,d^3p''\,d^3r\,d^3k'\,d^3p'\,d^3q'\,d^3k\,d^3p\ |q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)\rangle\ \langle q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)|\ S\ |q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\,g^{a'}_{i'}(q')\rangle \times$$
$$\frac{\langle q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\,g^{a'}_{i'}(q')|\,H_{qg}\,|q^{\alpha}_{\lambda_1}(k)\,\bar{q}^{\beta}_{\lambda_2}(p)\rangle}{E_{\bar{q}}(p')+E_q(k')+E_g(q')-E_\gamma} \times \frac{\langle q^{\alpha}_{\lambda_1}(k)\,\bar{q}^{\beta}_{\lambda_2}(p)|\,H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p)+E_q(k)-E_\gamma(q,Q)} \tag{2.31}$$

$$|\gamma^i(w)\rangle^{(3)}_{q\bar{q}g} = \frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ |q^\alpha_{\lambda_1}(\zeta q^+,x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y)g^a_m(\xi q^+,z)\rangle\ \frac{\eta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\times$$
$$\left[\tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}](\tilde{R},X)\times\left[U^{ab}(z)\,V(x)\,t^b\,V^\dagger(y)-t^a\right]_{\alpha\beta}\right] \tag{2.32}$$

with

$$\tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}] = \sum_\lambda\Big[\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\tilde{R}^j X^n\,\tilde{\mathcal{G}}^{(11)} + i\xi\,\tilde{\omega}_f\mu^m_{\lambda_1\lambda}\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\tilde{R}^j\tilde{\mathcal{G}}^{(10)}$$
$$- im_f\,\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,\mu^i_{\lambda\lambda_2}\,X^n\tilde{\mathcal{G}}^{(01)} + m_f\xi\,\tilde{\omega}_f\mu^m_{\lambda_1\lambda}\mu^i_{\lambda\lambda_2}\tilde{\mathcal{G}}^{(00)}\Big] \tag{2.33}$$

where

$$\tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R},X) = \frac{\tilde{a}^\beta}{2\,\eta(1-\eta)}\int_0^1\frac{dx}{x^{1+\beta}}\left(\frac{\tilde{A}_x}{\tilde{\Delta}_x}\right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{\tilde{A}_x\tilde{\Delta}_x}), \alpha,\beta\in\{0,1\}$$
$$\tilde{A}_x = \tilde{\epsilon}_f^2 + \frac{x\,\tilde{\omega}_f^2}{\tilde{a}} = \eta\,(1-\eta)\,Q^2 + m_f^2\left(1+\frac{x\eta\xi}{\zeta}\right)$$
$$\tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a}\,X^2}{x}$$

Explicitly

$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\,\eta\,(1-\eta)}\int_0^1\frac{dx}{x^2}\,\frac{\tilde{A}_x}{\tilde{\Delta}_x}\,K_2\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{2.34}$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\,\eta\,(1-\eta)}\int_0^1\frac{dx}{x}\sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}}\,K_1\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{2.35}$$

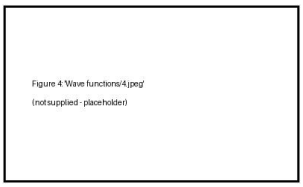
$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\,\eta\,(1-\eta)}\int_0^1\frac{dx}{x^2}\sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}}\,K_1\left(\sqrt{\tilde{A}_x\,\tilde{\Delta}_x}\right) \tag{2.36}$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\,\eta\,(1-\eta)}\int_0^1\frac{dx}{x}\,K_0\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{2.37}$$

5. Quark emits a gluon after the shock wave

$$|\gamma\rangle^{(4)}_{q\bar{q}g} = |q_2\bar{q}_2g_2\rangle\ \langle q_2\bar{q}_2g_2|\ S\ |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1|\,H_{qg}\,|q\bar{q}\rangle}{E_{\bar{q}_1}+E_{q_1}-E_{q_2}-E_{\bar{q}_2}-E_{g_2}} \times \frac{\langle q\bar{q}|\,H_{\gamma\to q\bar{q}}\,|\gamma\rangle}{E_q+E_{\bar{q}}-E_\gamma} \tag{2.38}$$

$$|\gamma\rangle^{(4)}_{q\bar{q}g} = \int d^3k''\,d^3p''\,d^3r\,d^3k'\,d^3p'\,d^3k\,d^3p\ |q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)\rangle\,\frac{\langle q^{\alpha''}_{\lambda_1''}(k'')\,\bar{q}^{\beta''}_{\lambda_2''}(p'')\,g^b_j(r)|\,H_{qg}\,|q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\rangle}{E_q(k')+E_{\bar{q}}(p')-E_q(k'')-E_{\bar{q}}(p'')-E_g(r)}\times$$
$$\langle q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')|\,S\,|q^{\alpha}_{\lambda_1}(k)\,\bar{q}^{\beta}_{\lambda_2}(p)\rangle \times \frac{\langle q^{\alpha}_{\lambda_1}(k)\,\bar{q}^{\beta}_{\lambda_2}(p)|\,H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p)+E_q(k)-E_\gamma(q,Q)} \tag{2.39}$$



Substituting the relations we get

$$|\gamma^i(w)\rangle^{(4)}_{q\bar{q}g} = -\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ \langle q^\alpha_{\lambda_1}(\zeta q^+,x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y)g^a_m(\xi q^+,z)\rangle\ \frac{\eta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\times$$
$$\tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{A}}](\tilde{R},X)\times\left[t^aV(x')V^\dagger(y)-t^a\right]_{\alpha\beta} \tag{2.40}$$

where

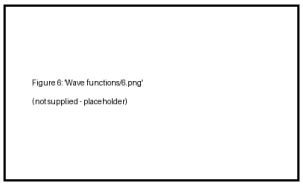
$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f\,K_1(\tilde{\epsilon}_f\,\tilde{R})}{\eta\,(1-\eta)\,\tilde{R}}\,\frac{\tilde{\omega}_f\,K_1(\tilde{\omega}_f\,X)}{X} \tag{2.41}$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f\,K_1(\tilde{\epsilon}_f\tilde{R})}{\eta\,(1-\eta)\,\tilde{R}}\,K_0(\tilde{\omega}_f\,X) \tag{2.42}$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f\tilde{R})}{\eta\,(1-\eta)}\,\frac{\tilde{\omega}_f\,K_1(\tilde{\omega}_f\,X)}{X} \tag{2.43}$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f\,\tilde{R})}{\eta\,(1-\eta)}\,K_0(\tilde{\omega}_f\,X) \tag{2.44}$$

6. Instantaneous contribution (Quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle^{(3\,\text{inst.})}_{q\bar{q}g} = |q\bar{q}g\rangle\ \langle q\bar{q}g|\ S\ |q\bar{q}g\rangle\ \frac{\langle q\bar{q}g|\ H_{\gamma\to q\bar{q}g}\,|\gamma\rangle}{E_{q\bar{q}g}-E_\gamma} \tag{2.45}$$

Using the transformations from anti quark to quark with the insertions of Wilson lines, we get

$$|\gamma\rangle^{(3\,\text{inst.})}_{q\bar{q}g} = -\frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ \langle\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y),g^a_m(\xi\,q^+,z),q^\alpha_{\lambda_1}(\zeta\,q^+,x)\rangle\ \frac{\zeta\,\eta\,\sqrt{\xi}}{(1-\eta)}\,\delta^{(2)}(w-c)\ \left(\delta^{im}-2i\,\epsilon^{im}\lambda_2\right)\ \delta_{\lambda_1\lambda_2}$$
$$\frac{E_f\,K_1(E_f\mathcal{D})}{\mathcal{D}}\ \left[U^{ab}(z)V(x)t^bV^\dagger(y)-t^a\right]_{\alpha\beta} \tag{2.46}$$

Now, we will write the quark emissions as how we wrote the anti quark section. We will use the effective vertex formalism.

$$|\gamma^i(w)\rangle_{q\bar{q}g} = \frac{e\,e_f\,g\,q^+}{2\,(2\pi)^4}\int_{x,y,z}\int_0^1 d\zeta\int_0^{1-\zeta} d\xi\ \langle q^\alpha_{\lambda_1}(\zeta q^+,x)\bar{q}^\beta_{\lambda_2}(\eta\,q^+,y)g^a_m(\xi q^+,z)\rangle\ \frac{\eta}{\sqrt{\xi}}\,\delta^{(2)}(w-c)\times$$
$$\left[\tilde{\mathcal{M}}^{im}_{1\lambda_1\lambda_2}(x,y,z,\eta,\xi)\times\ \left[U^{ab}(z)\,V(x)\,t^b\,V^\dagger(y)-t^a\right]_{\alpha\beta}-\tilde{\mathcal{M}}^{im}_{2\lambda_1\lambda_2}\left[t^a\,V(x')\,V^\dagger(y)-t^a\right]_{\alpha\beta}\right] \tag{2.47}$$

with the effective vertex

$$\tilde{\mathcal{M}}^{im}_{1\lambda_1\lambda_2}(x,y,z,\eta,\xi) = \tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}](\tilde{R},X) - \delta_{\lambda_1\lambda_2}\,\frac{\zeta\,\xi}{(\xi+\zeta)}\,(\delta^{im}-2i\,\epsilon^{im}\lambda_1)\,\mathcal{G}_E \tag{2.48}$$

$$\tilde{\mathcal{M}}^{im}_{2\lambda_1\lambda_2}(x,y,z,\eta,\xi) = \tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{A}}](\tilde{R},X) \tag{2.49}$$

3 Trijet production cross section

3.1 Direct contributions - Anti quark contribution

$$\frac{d^3N^{\rm direct}}{d^3k_1\,d^3k_2\,d^3k_3} = \frac{e^2\,g^2\,\zeta^2\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+\,2}}\,\delta\left(q^+ - k_1^+ - k_2^+ - k_3^+\right)\int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i\,k_1\cdot(x-\bar{x})}\,e^{-i\,k_2\cdot(y-\bar{y})}\,e^{-i\,k_3\cdot(z-\bar{z})}\,\left(\sum_f e_f^2\right)$$
$$\left[\mathcal{H}[\mathcal{G};\bar{\mathcal{G}}]\,\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) - \mathcal{H}[\mathcal{A};\bar{\mathcal{G}}]\,\mathcal{W}(x,y',y',\bar{x},\bar{y},\bar{z}) - \mathcal{H}[\mathcal{G};\bar{\mathcal{A}}]\,\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}') + \mathcal{H}[\mathcal{A};\bar{\mathcal{A}}]\,\mathcal{W}(x,y',y',\bar{x},\bar{y}',\bar{y}')\right] \tag{3.1}$$

where

$$\mathcal{H}[F;\bar{F}] = \sum_{\lambda_1,\lambda_2}\sum_{l,r}\mathcal{M}^{lr}_{\lambda_1\lambda_2}[F](x,y,z)\,\mathcal{M}^{lr\,*}_{\lambda_1\lambda_2}[\bar{F}](\bar{x},\bar{y},\bar{z}) \tag{3.2}$$

- Regular × Regular

$$\begin{aligned}\mathcal{H}_{\rm reg.\times reg.}[F;\bar{F}] = {} & R^i\,Y^m\,\bar{R}^j\,\bar{Y}^n\left\{8\,\left[\eta^2+(1-\zeta)^2\right]\,\left[\zeta^2+(1-\zeta)^2\right]\,\delta^{ij}\delta^{mn} - 8\xi(2\zeta-1)\,(2(1-\zeta)-\xi)\epsilon^{ij}\,\epsilon^{mn}\right\}F^{(11)}\,\bar{F}^{(11)}\\ & + 8\,m_f^2\,\left[\eta^2+(1-\zeta)^2\right]\,(Y\cdot\bar{Y})\,F^{(01)}\,\bar{F}^{(01)} + 8\,\xi^2\,\omega_f^2\,\left[\zeta^2+(1-\zeta)^2\right]\,(R\cdot\bar{R})\,F^{(10)}\,\bar{F}^{(10)}\\ & + 8\,m_f^2\,\xi^2\,\omega_f^2\,F^{(00)}\,\bar{F}^{(00)}\\ & - 8\zeta\eta\,m_f\,\xi\,\omega_f\,\left[(R\cdot Y)\,F^{(11)}\,\bar{F}^{(00)} + (\bar{R}\cdot\bar{Y})\,F^{(00)}\,\bar{F}^{(11)} + (R\cdot\bar{Y})\,F^{(10)}\,\bar{F}^{(01)} + (Y\cdot\bar{R})\,F^{(01)}\,\bar{F}^{(10)}\right]\end{aligned} \tag{3.3}$$

- Regular × Instantaneous + h.c.

$$\begin{aligned}\mathcal{H}_{\rm reg.\times inst.}[F;\bar{F}] = {} & -\frac{8\zeta\xi\eta^2}{(1-\zeta)}\,\left[(R\cdot Y)\,F^{(11)}\,\bar{\mathcal{G}}_E + (\bar{R}\cdot\bar{Y})\mathcal{G}_E\,\bar{F}^{(11)}\right]\\ & + \frac{8\xi^2\,\eta\,m_f\,\omega_f}{(1-\zeta)}\,\left[F^{(00)}\,\bar{\mathcal{G}}_E + \mathcal{G}_E\,\bar{F}^{(00)}\right]\end{aligned} \tag{3.4}$$

- Instantaneous × Instantaneous

$$\mathcal{H}_{\rm inst.\times inst.}[F;\bar{F}] = 8\,\left[\frac{\xi\,\eta}{(1-\zeta)}\right]^2\,\mathcal{G}_E\,\bar{\mathcal{G}}_E \tag{3.5}$$

where:

$$\mathcal{M}^{im}_{\lambda_1\lambda_2}[\mathcal{G}] = \mathcal{M}^{im}_{1\,\lambda_1\lambda_2}[\mathcal{G}] = \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}](R,Y) - \delta_{\lambda_1\lambda_2}\,\frac{\xi\,\eta}{(1-\zeta)}\,\left(\delta^{im} + 2i\epsilon^{im}\,\lambda_1\right)\,\mathcal{G}_E \tag{3.6}$$

$$\mathcal{M}^{im}_{\lambda_1\lambda_2}[\mathcal{A}] = \mathcal{M}^{im}_{2\lambda_1\lambda_2}[\mathcal{A}] = \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}](R,Y) \tag{3.7}$$

$$\begin{aligned}\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}] = \sum_\lambda \Big[& \varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,R^j\,Y^n\,\mathcal{G}^{(11)} + i\,\xi\,\omega_f\,\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\mu^m_{\lambda\lambda_2}\,R^j\,\mathcal{G}^{(10)}\\ & - i\,m_f\,\mu^i_{\lambda_1\lambda}\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,Y^n\,\mathcal{G}^{(01)} + m_f\,\xi\,\omega_f\,\mu^i_{\lambda_1\lambda}\,\mu^m_{\lambda\lambda_2}\,\mathcal{G}^{(00)}\Big]\end{aligned} \tag{3.8}$$

Explicitly:

$$\mathcal{G}^{(\alpha\beta)}(R,Y) = \frac{a^\beta}{2\,\zeta(1-\zeta)}\,\int_0^1\,\frac{dx}{x^{1+\beta}}\,\left(\frac{A_x}{\Delta_x}\right)^{\frac{(\alpha+\beta)}{2}}\,K_{\alpha+\beta}(\sqrt{A_x\Delta_x}), \alpha,\beta\in\{0,1\}$$

$$A_x = \epsilon_f^2 + \frac{x\,\omega_f^2}{a} = \zeta\,(1-\zeta)\,Q^2 + m_f^2\,\left(1+\frac{x\zeta\xi}{\eta}\right)$$

$$\Delta_x = R^2 + \frac{a\,Y^2}{x}$$

$$\mathcal{G}^{(11)} = \frac{a}{2\,\zeta\,(1-\zeta)}\,\int_0^1\,\frac{dx}{x^2}\,\frac{A_x}{\Delta_x}\,K_2\left(\sqrt{A_x\Delta_x}\right) \tag{3.9}$$

$$\mathcal{G}^{(10)} = \frac{1}{2\,\zeta\,(1-\zeta)}\,\int_0^1\,\frac{dx}{x}\,\sqrt{\frac{A_x}{\Delta_x}}\,K_1\left(\sqrt{A_x\Delta_x}\right) \tag{3.10}$$

$$\mathcal{G}^{(01)} = \frac{a}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}}\, K_1\left(\sqrt{A_x\,\Delta_x}\right) \tag{3.11}$$

$$\mathcal{G}^{(00)} = \frac{1}{2\,\zeta\,(1-\zeta)} \int_0^1 \frac{dx}{x}\, K_0\left(\sqrt{A_x \Delta_x}\right) \tag{3.12}$$

and

$$\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}] = \sum_{\lambda} \Big[\varphi^{ij}_{\lambda_1\lambda}(\zeta)\, \tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\, R^j\, Y^n\, \mathcal{A}^{(11)} + i\,\xi\,\omega_f\, \varphi^{ij}_{\lambda_1\lambda}(\zeta)\, \mu^m_{\lambda\lambda_2}\, R^j\, \mathcal{A}^{(10)}$$
$$- i\, m_f\, \mu^i_{\lambda_1\lambda}\, \tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\, Y^n\, \mathcal{A}^{(01)} + m_f\,\xi\,\omega_f\, \mu^i_{\lambda_1\lambda}\, \mu^m_{\lambda\lambda_2}\, \mathcal{A}^{(00)} \Big] \tag{3.13}$$

Explicitly:

$$\mathcal{A}^{(11)} = \frac{\epsilon_f\, K_1(\epsilon_f R)}{\zeta(1-\zeta)\, R}\, \frac{\omega_f\, K_1(\omega_f\, Y)}{Y}$$

$$\mathcal{A}^{(10)} = \frac{\epsilon_f\, K_1(\epsilon_f R)}{\zeta(1-\zeta)\, R}\, K_0(\omega_f Y)$$

$$\mathcal{A}^{(01)} = \frac{K_0(\epsilon_f R)}{\zeta\,(1-\zeta)}\, \frac{\omega_f\, K_1(\omega_f Y)}{Y}$$

$$\mathcal{A}^{(00)} = \frac{K_0(\epsilon_f R)}{\zeta\,(1-\zeta)}\, K_0(\omega_f Y)$$

Finally:

$$\mathcal{G}_E = \frac{E_f\, K_1(E_f \mathcal{D})}{\mathcal{D}}$$

where:

$$\mathcal{D}^2(x,y,z,\zeta,\xi) = \zeta\,(1-\zeta)\, R^2 + \frac{\xi\,\eta\, Y^2}{(1-\zeta)} = \zeta\,\eta\,(x-y)^2 + \zeta\,\xi\,(x-z)^2 + \eta\,\xi\,(y-z)^2$$

$$\mu^i_{\lambda\lambda'} = \delta_{\lambda',-\lambda}\left[2\lambda\delta^{i1} - i\delta^{i2}\right] = \chi^\dagger_\lambda \sigma^3 \sigma^i \chi_{\lambda'}$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta\,(1-\zeta)\, Q^2 + m_f^2, \qquad \omega_f = \frac{\xi\, m_f}{(1-\zeta)}, \qquad E_f^2 = Q^2 + m_f^2\left(\frac{1}{\zeta} + \frac{1}{\eta}\right), \qquad a = \frac{\xi\,\eta}{\zeta\,(1-\zeta)^2}$$

$$\varphi^{ij}_{\lambda_1\lambda}(\zeta) = \delta_{\lambda_1\lambda}\left[(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1\right], \qquad \tau^{mn}_{\lambda\lambda_2}(\xi,\eta) = \delta_{\lambda\lambda_2}\,\left[(2(1-\zeta)-\xi)\delta^{mn} + 2i\,\xi\,\epsilon^{mn}\,\lambda\right]$$

and

$$c = \zeta\, x + \eta\, y + \xi\, z$$

is the centre of energy of trijet system. $\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z})$

$$\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y},\bar{z}) - S_{q\bar{q}g}(x,y,z) - S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) + 1 \tag{3.14}$$

  At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y},\bar{z}) = Q(x,z,\bar{z},\bar{x})\, Q(z,y,\bar{y},\bar{z})$$

$$S_{q\bar{q}g}(x,y,z) = S(x,z)\, S(z,y)$$

$$S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) = S(\bar{z},\bar{x})\, S(\bar{y},\bar{z})$$

$$\boxed{\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) = Q(x,z,\bar{z},\bar{x})\, Q(z,y,\bar{y},\bar{z}) - S(x,z)\, S(z,y) - S(\bar{z},\bar{x})\, S(\bar{y},\bar{z}) + 1}$$

- $\mathcal{W}(x,y',y',\bar{x},\bar{y},\bar{z})$

$$\mathcal{W}(x,y',y',\bar{x},\bar{y},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x,y',y',\bar{x},\bar{y},\bar{z}) - S_{q\bar{q}g}(x,y',y') - S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) + 1 \tag{3.15}$$

  At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y',y',\bar{x},\bar{y},\bar{z}) = Q(x,y',\bar{z},\bar{x})\, Q(y',y',\bar{y},\bar{z}) = Q(x,y',\bar{z},\bar{x})\, S(\bar{y},\bar{z})$$

$$S_{q\bar{q}g}(x,y',y') = S(x,y')\, S(y',y') = S(x,y')$$

$$S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) = S(\bar{z},\bar{x})\, S(\bar{y},\bar{z})$$

$$\boxed{\mathcal{W}(x,y',y',\bar{x},\bar{y},\bar{z}) = Q(x,y',\bar{z},\bar{x})\, S(\bar{y},\bar{z}) - S(x,y') - S(\bar{z},\bar{x})\, S(\bar{y},\bar{z}) + 1}$$

- $\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}')$

$$\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y}',\bar{y}') - S_{q\bar{q}g}(x,y,z) - S^{\dagger}_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') + 1 \tag{3.16}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,Q(z,y,\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,S(z,y)$$

$$S_{q\bar{q}g}(x,y,z) = S(x,z)\,S(z,y)$$

$$S^{\dagger}_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})\,S(\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})$$

$$\boxed{\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,S(z,y) - S(x,z)\,S(z,y) - S(\bar{y}',\bar{x}) + 1}$$

- $\mathcal{W}(x,y',y',\bar{x},\bar{y}',\bar{y}')$

$$\mathcal{W}(x,y',y',\bar{x},\bar{y}',\bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x,y',y',\bar{x},\bar{y}',\bar{y}') - S_{q\bar{q}g}(x,y',y') - S^{\dagger}_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') + 1 \tag{3.17}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y',y',\bar{x},\bar{y}',\bar{y}') = Q(x,y',\bar{y}',\bar{x})\,Q(y',y',\bar{y}',\bar{y}') = Q(x,y',\bar{y}',\bar{x})$$

$$S_{q\bar{q}g}(x,y',y') = S(x,y')\,S(y',y') = S(x,y')$$

$$S^{\dagger}_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})\,S(\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})$$

$$\boxed{\mathcal{W}(x,y',y',\bar{x},\bar{y}',\bar{y}') = Q(x,y',\bar{y}',\bar{x}) - S(x,y') - S(\bar{y}',\bar{x}) + 1}$$

3.2 Direct contribution - Quark contributions

The following transformations are made from the anti quark sector to quark sector:

$$\zeta \to \eta, R \to \tilde{R}, Y \to X, x \leftrightarrow y$$

and

$$\epsilon_f \to \tilde{\epsilon}_f, \omega_f \to \tilde{\omega}_f, \mathcal{G} \to \tilde{\mathcal{G}}, \mathcal{A} \to \tilde{\mathcal{A}}$$

$$\frac{d^3N^{\rm direct}}{d^3k_1\,d^3k_2\,d^3k_3} = \frac{e^2\,g^2\,\eta^2\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+\,2}}\,\delta\left(q^+ - k_1^+ - k_2^+ - k_3^+\right)\int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i\,k_1\cdot(x-\bar{x})}\,e^{-i\,k_2\cdot(y-\bar{y})}\,e^{-i\,k_3\cdot(z-\bar{z})}\,\left(\sum_f e_f^2\right)$$
$$\left[\tilde{\mathcal{H}}[\tilde{\mathcal{G}};\bar{\tilde{\mathcal{G}}}]\,\mathcal{W}(y,x,z,\bar{y},\bar{x},\bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{A}};\bar{\tilde{\mathcal{G}}}]\,\mathcal{W}(y,x',x',\bar{y},\bar{x},\bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{G}};\bar{\tilde{\mathcal{A}}}]\,\mathcal{W}(y,x,z,\bar{y},\bar{x}',\bar{x}') + \tilde{\mathcal{H}}[\tilde{\mathcal{A}};\bar{\tilde{\mathcal{A}}}]\,\mathcal{W}(y,x',x',\bar{y},\bar{x}',\bar{x}')\right] \tag{3.18}$$

where:

$$\tilde{\mathcal{H}}[F;\bar{F}] = \sum_{\lambda_1,\lambda_2}\sum_{l,r}\tilde{\mathcal{M}}^{lr}_{\lambda_1\lambda_2}[F](x,y,z)\,\tilde{\mathcal{M}}^{lr\,*}_{\lambda_1\lambda_2}[\bar{F}](\bar{x},\bar{y},\bar{z}) \tag{3.19}$$

- Regular $\times$ Regular

$$\tilde{\mathcal{H}}_{\rm reg.\times reg.} = \tilde{R}^i\,X^m\,\bar{\tilde{R}}^j\,\bar{X}^n\left\{8\,\left[\zeta^2+(1-\eta)^2\right]\,\left[\eta^2+(1-\eta)^2\right]\,\delta^{ij}\delta^{mn} - 8\xi\,(2\eta-1)\,(2(1-\eta)-\xi)\,\epsilon^{ij}\,\epsilon^{mn}\right\}F^{(11)}\,\bar{F}^{(11)}$$
$$+ 8m_f^2\left[\zeta^2+(1-\eta)^2\right]\,(X\cdot\bar{X})\,F^{(01)}\,\bar{F}^{(01)} + 8\xi^2\tilde{\omega}_f^2\left[\eta^2+(1-\eta)^2\right]\,(\tilde{R}\cdot\bar{\tilde{R}})\,F^{(10)}\bar{F}^{(10)} + 8m_f^2\xi^2\tilde{\omega}_f^2\,F^{(00)}\bar{F}^{(00)}$$
$$+ 8\zeta\,\eta\,m_f\,\xi\,\tilde{\omega}_f\,\left[(\tilde{R}\cdot X)\,F^{(11)}\bar{F}^{(00)} + (\bar{\tilde{R}}\cdot\bar{X})\,F^{(00)}\bar{F}^{(11)} + (\tilde{R}\cdot\bar{X})\,F^{(10)}\bar{F}^{(01)} + (X\cdot\bar{\tilde{R}})\,F^{(01)}\bar{F}^{(10)}\right] \tag{3.20}$$

- Regular $\times$ Instantaneous + h.c.

$$\tilde{\mathcal{H}}_{\rm reg.\times inst.} = \frac{8\,\zeta^2\,\xi\,\eta}{(1-\eta)}\,\left[(\tilde{R}\cdot X)\,F^{(11)}\,\bar{\mathcal{G}}_E + (\bar{\tilde{R}}\cdot\bar{X})\,\mathcal{G}_E\,\bar{F}^{(11)}\right]$$
$$+ \frac{8\,\xi^2\,\zeta\,m_f\,\tilde{\omega}_f}{(1-\eta)}\,\left[F^{(00)}\,\bar{\mathcal{G}}_E + \mathcal{G}_E\,\bar{F}^{(00)}\right] \tag{3.21}$$

- Instantaneous $\times$ Instantaneous

$$\tilde{\mathcal{H}}_{\rm inst.\times inst.} = 8\,\left[\frac{\xi\,\zeta}{(1-\eta)}\right]^2\,\mathcal{G}_E\,\bar{\mathcal{G}}_E \tag{3.22}$$

where:

$$\tilde{\mathcal{M}}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}] = \tilde{\mathcal{M}}^{im}_{1\lambda_1\lambda_2}[\tilde{\mathcal{G}}] = \tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}](\tilde{R}, X) - \delta_{\lambda_1\lambda_2}\,\frac{\zeta\,\xi}{(\xi+\zeta)}\,(\delta^{im} - 2i\,\epsilon^{im}\lambda_1)\,\mathcal{G}_E \tag{3.23}$$

$$\tilde{\mathcal{M}}^{im}_{2\lambda_1\lambda_2}[\tilde{\mathcal{A}}] = \tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{A}}](\tilde{R}, X) \tag{3.24}$$

with

$$\tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{G}}] = \sum_{\lambda}\Bigg[\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\tilde{R}^j X^n\,\tilde{\mathcal{G}}^{(11)} + i\xi\,\tilde{\omega}_f\mu^m_{\lambda_1\lambda}\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\tilde{R}^j\tilde{\mathcal{G}}^{(10)}$$
$$- im_f\,\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,\mu^i_{\lambda\lambda_2}\,X^n\tilde{\mathcal{G}}^{(01)} + m_f\xi\,\tilde{\omega}_f\mu^m_{\lambda_1\lambda}\mu^i_{\lambda\lambda_2}\tilde{\mathcal{G}}^{(00)}\Bigg] \tag{3.25}$$

where

$$\tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X) = \frac{\tilde{a}^\beta}{2\,\eta(1-\eta)}\int_0^1 \frac{dx}{x^{1+\beta}}\left(\frac{\tilde{A}_x}{\tilde{\Delta}_x}\right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{\tilde{A}_x\tilde{\Delta}_x}), \alpha,\beta\in\{0,1\}$$

$$\tilde{A}_x = \tilde{\epsilon}_f^2 + \frac{x\,\tilde{\omega}_f^2}{\tilde{a}} = \eta\,(1-\eta)\,Q^2 + m_f^2\,\left(1+\frac{x\eta\xi}{\zeta}\right)$$

$$\tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a}\,X^2}{x}$$

Explicitly

$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\,\eta\,(1-\eta)}\int_0^1 \frac{dx}{x^2}\,\frac{\tilde{A}_x}{\tilde{\Delta}_x}\,K_2\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{3.26}$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\,\eta\,(1-\eta)}\int_0^1 \frac{dx}{x}\,\sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}}\,K_1\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{3.27}$$

$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\,\eta\,(1-\eta)}\int_0^1 \frac{dx}{x^2}\,\sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}}\,K_1\left(\sqrt{\tilde{A}_x\,\tilde{\Delta}_x}\right) \tag{3.28}$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\,\eta\,(1-\eta)}\int_0^1 \frac{dx}{x}\,K_0\left(\sqrt{\tilde{A}_x\tilde{\Delta}_x}\right) \tag{3.29}$$

and

$$\tilde{\Psi}^{im}_{\lambda_1\lambda_2}[\tilde{\mathcal{A}}] = \sum_{\lambda}\Bigg[\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\tilde{R}^j X^n\,\tilde{\mathcal{A}}^{(11)} + i\xi\,\tilde{\omega}_f\,\mu^m_{\lambda_1\lambda}\,\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\tilde{R}^j\tilde{\mathcal{A}}^{(10)}$$
$$- i\,m_f\,\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\mu^i_{\lambda\lambda_2}\,X^n\,\tilde{\mathcal{A}}^{(01)} + m_f\,\xi\,\tilde{\omega}_f\,\mu^m_{\lambda_1\lambda}\,\mu^i_{\lambda\lambda_2}\,\tilde{\mathcal{A}}^{(00)}\Bigg] \tag{3.30}$$

Explicitly:

$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f\,K_1(\tilde{\epsilon}_f\,\tilde{R})}{\eta\,(1-\eta)\,\tilde{R}}\,\frac{\tilde{\omega}_f\,K_1(\tilde{\omega}_f\,X)}{X} \tag{3.31}$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f\,K_1(\tilde{\epsilon}_f\tilde{R})}{\eta\,(1-\eta)\,\tilde{R}}\,K_0(\tilde{\omega}_f\,X) \tag{3.32}$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f\tilde{R})}{\eta\,(1-\eta)}\,\frac{\tilde{\omega}_f\,K_1(\tilde{\omega}_f\,X)}{X} \tag{3.33}$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f\,\tilde{R})}{\eta\,(1-\eta)}\,K_0(\tilde{\omega}_f\,X) \tag{3.34}$$

and

- $\xi\,\omega_f\,\mu^m_{\lambda\lambda_2} \to \xi\,\tilde{\omega}_f\,\mu^m_{\lambda_1\lambda}$ with $\tilde{\omega}_f = \frac{\xi\,m_f}{(1-\eta)}$

- $\epsilon_f^2 \to \tilde{\epsilon}_f^2 = \eta(1-\eta)\,Q^2 + m_f^2$, $a \to \tilde{a} = \frac{\xi\,\zeta}{\eta\,(1-\eta)^2}$
- $A_x \to \tilde{A}_x = \eta\,(1-\eta)\,Q^2 + m_f^2\left(1 + \frac{x\eta\xi}{\zeta}\right)$ and $\Delta_x \to \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a}\,X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R,Y) \to \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R},X)$ and $\mathcal{A}^{(\alpha,\beta)}(R,Y) \to \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R},X)$ also, $\zeta\,(1-\zeta) \to \eta\,(1-\eta)$

$\mathcal{W}(y,x,z,\bar{y},\bar{x},\bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(y,x,z,\bar{y},\bar{x},\bar{z})$

$$\mathcal{W}(y,x,z,\bar{y},\bar{x},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(y,x,z,\bar{y},\bar{x},\bar{z}) - S_{q\bar{q}g}(y,x,z) - S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x},\bar{z}) + 1 \tag{3.35}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y,x,z,\bar{y},\bar{x},\bar{z}) = Q(y,z,\bar{z},\bar{y})\,Q(z,x,\bar{x},\bar{z})$$
$$S_{q\bar{q}g}(y,x,z) = S(y,z)\,S(z,x)$$
$$S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x},\bar{z}) = S(\bar{z},\bar{y})\,S(\bar{x},\bar{z})$$

$$\boxed{\mathcal{W}(y,x,z,\bar{y},\bar{x},\bar{z}) = Q(y,z,\bar{z},\bar{y})\,Q(z,x,\bar{x},\bar{z}) - S(y,z)\,S(z,x) - S(\bar{z},\bar{y})\,S(\bar{x},\bar{z}) + 1}$$

- $\mathcal{W}(y,x',x',\bar{y},\bar{x},\bar{z})$

$$\mathcal{W}(y,x',x',\bar{y},\bar{x},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(y,x',x',\bar{y},\bar{x},\bar{z}) - S_{q\bar{q}g}(y,x',x') - S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x},\bar{z}) + 1 \tag{3.36}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y,x',x',\bar{y},\bar{x},\bar{z}) = Q(y,x',\bar{z},\bar{y})\,Q(x',x',\bar{x},\bar{z}) = Q(y,x',\bar{z},\bar{y})\,S(\bar{x},\bar{z})$$
$$S_{q\bar{q}g}(y,x',x') = S(y,x')\,S(x',x') = S(y,x')$$
$$S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x},\bar{z}) = S(\bar{z},\bar{y})\,S(\bar{x},\bar{z})$$

$$\boxed{\mathcal{W}(y,x',x',\bar{y},\bar{x},\bar{z}) = Q(y,x',\bar{z},\bar{y})\,S(\bar{x},\bar{z}) - S(y,x') - S(\bar{z},\bar{y})\,S(\bar{x},\bar{z}) + 1}$$

- $\mathcal{W}(y,x,z,\bar{y},\bar{x}',\bar{x}')$

$$\mathcal{W}(y,x,z,\bar{y},\bar{x}',\bar{x}') = S_{q\bar{q}g\,q\bar{q}g}(y,x,z,\bar{y},\bar{x}',\bar{x}') - S_{q\bar{q}g}(y,x,z) - S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x}',\bar{x}') + 1 \tag{3.37}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y,x,z,\bar{y},\bar{x}',\bar{x}') = Q(y,z,\bar{x}',\bar{y})\,Q(z,x,\bar{x}',\bar{x}') = Q(y,z,\bar{x}',\bar{y})\,S(z,x)$$
$$S_{q\bar{q}g}(y,x,z) = S(y,z)\,S(z,x)$$
$$S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x}',\bar{x}') = S(\bar{x}',\bar{y})\,S(\bar{x}',\bar{x}') = S(\bar{x}',\bar{y})$$

$$\boxed{\mathcal{W}(y,x,z,\bar{y},\bar{x}',\bar{x}') = Q(y,z,\bar{x}',\bar{y})\,S(z,x) - S(y,z)\,S(z,x) - S(\bar{x}',\bar{y}) + 1}$$

- $\mathcal{W}(y,x',x',\bar{y},\bar{x}',\bar{x}')$

$$\mathcal{W}(y,x',x',\bar{y},\bar{x}',\bar{x}') = S_{q\bar{q}g\,q\bar{q}g}(y,x',x',\bar{y},\bar{x}',\bar{x}') - S_{q\bar{q}g}(y,x',x') - S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x}',\bar{x}') + 1 \tag{3.38}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y,x',x',\bar{y},\bar{x}',\bar{x}') = Q(y,x',\bar{x}',\bar{y})\,Q(x',x',\bar{x}',\bar{x}') = Q(y,x',\bar{x}',\bar{y})$$
$$S_{q\bar{q}g}(y,x',x') = S(y,x')\,S(x',x') = S(y,x')$$
$$S^{\dagger}_{q\bar{q}g}(\bar{y},\bar{x}',\bar{x}') = S(\bar{x}',\bar{y})\,S(\bar{x}',\bar{x}') = S(\bar{x}',\bar{y})$$

$$\boxed{\mathcal{W}(y,x',x',\bar{y},\bar{x}',\bar{x}') = Q(y,x',\bar{x}',\bar{y}) - S(y,x') - S(\bar{x}',\bar{y}) + 1}$$

3.3 Interference contributions

$$\frac{d^3N^{\rm cross}}{d^3k_1\,d^3k_2\,d^3k_3} = -\frac{e^2\,g^2\,\eta\,\zeta\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+\,2}}\,\delta\left(q^+ - k_1^+ - k_2^+ - k_3^+\right)\times\,2\,\Re\int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i\,k_1\cdot(x-\bar{x})}\,e^{-i\,k_2\cdot(y-\bar{y})}\,e^{-i\,k_3\cdot(z-\bar{z})}\left(\sum_f e_f^2\right)$$
$$\Bigg[\mathcal{H}^x[\tilde{\mathcal{G}};\bar{\mathcal{G}}]\,\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) - \mathcal{H}^x[\tilde{\mathcal{A}};\bar{\mathcal{G}}]\,\mathcal{W}(x',y,x',\bar{x},\bar{y},\bar{z}) - \mathcal{H}^x[\tilde{\mathcal{G}};\bar{\mathcal{A}}]\,\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}')$$
$$+\,\mathcal{H}^x[\tilde{\mathcal{A}};\bar{\mathcal{A}}]\,\mathcal{W}(x',y,x',\bar{x},\bar{y}',\bar{y}')\Bigg] \tag{3.39}$$

where

$$\mathcal{H}^x[\tilde{F};\bar{F}] = \sum_{\lambda_1,\lambda_2}\sum_{l,r}\tilde{\mathcal{M}}^{lr}_{\lambda_1\lambda_2}[\tilde{F}](x,y,z)\,\mathcal{M}^{lr\,*}_{\lambda_1\lambda_2}[\bar{F}](\bar{x},\bar{y},\bar{z}) \tag{3.40}$$

- Regular × Regular

$$\mathcal{H}^x_{\rm reg.\times reg.} = \tilde{R}^i\,X^m\,\bar{R}^j\,\bar{Y}^n\left\{8[1-\xi-2\zeta\eta]\,[\xi(1-\xi)+2\zeta\eta]\,\delta^{ij}\delta^{mn} - 8\xi[1-2\zeta-\xi]^2\,\epsilon^{ij}\epsilon^{mn}\right\}\tilde{F}^{(11)}\,\bar{F}^{(11)}$$
$$+\,8\,m_f^2\,(\xi+2\zeta\eta)\,(X\cdot\bar{Y})\,\tilde{F}^{(01)}\bar{F}^{(01)} - 8(\xi+2\eta\zeta)\xi^2\,\tilde{\omega}_f\omega_f\,(\tilde{R}\cdot\bar{R})\,\tilde{F}^{(10)}\bar{F}^{(10)} - 8(\zeta+\xi)^2\,m_f\,\xi\,\omega_f\,(\tilde{R}\cdot X)\,\tilde{F}^{(11)}\bar{F}^{(00)}$$
$$+\,8(1-\zeta)^2\,m_f\,\xi\,\tilde{\omega}_f\,(\bar{R}\cdot\bar{Y})\,\tilde{F}^{(00)}\bar{F}^{(11)} + 8\eta^2\,m_f\,\xi\,\tilde{\omega}_f\,(\tilde{R}\cdot\bar{Y})\,\tilde{F}^{(10)}\,\bar{F}^{(01)} - 8\zeta^2\,m_f\,\xi\,\omega_f\,(X\cdot\bar{R})\,\tilde{F}^{(01)}\bar{F}^{(10)} \tag{3.41}$$

- Regular × Instantaneous + h.c.

$$\mathcal{H}^x_{\rm reg.\times inst.} = -\frac{8\xi\,\eta(\xi+\zeta)^2}{(1-\zeta)}\,(\tilde{R}\cdot X)\,\tilde{F}^{(11)}\,\bar{\mathcal{G}}_E + \frac{8\,\xi\,\zeta\,(1-\zeta)^2}{(\xi+\zeta)}\,(\bar{R}\cdot\bar{Y})\,\mathcal{G}_E\,\bar{F}^{(11)} \tag{3.42}$$

- Instantaneous × Instantaneous

$$\mathcal{H}^x_{\rm inst.\times inst.} = \frac{\zeta\,\xi}{(\zeta+\xi)}\,\frac{\xi\,\eta}{(1-\zeta)}\mathcal{G}_E\,\bar{\mathcal{G}_E}\,\sum_{\lambda_1}\sum_{l,r}\left(\delta^{lr} - 2i\epsilon^{lr}\lambda_1\right)\left(\delta^{lr} + 2i\epsilon^{lr}\lambda_1\right)^* = 0 \tag{3.43}$$

$\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z})$

$$\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y},\bar{z}) - S_{q\bar{q}g}(x,y,z) - S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) + 1 \tag{3.44}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y},\bar{z}) = Q(x,z,\bar{z},\bar{x})\,Q(z,y,\bar{y},\bar{z})$$
$$S_{q\bar{q}g}(x,y,z) = S(x,z)\,S(z,y)$$
$$S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) = S(\bar{z},\bar{x})\,S(\bar{y},\bar{z})$$

$$\boxed{\mathcal{W}(x,y,z,\bar{x},\bar{y},\bar{z}) = Q(x,z,\bar{z},\bar{x})\,Q(z,y,\bar{y},\bar{z}) - S(x,z)\,S(z,y) - S(\bar{z},\bar{x})\,S(\bar{y},\bar{z}) + 1}$$

- $\mathcal{W}(x',y,x',\bar{x},\bar{y},\bar{z})$

$$\mathcal{W}(x',y,x',\bar{x},\bar{y},\bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x',y,x',\bar{x},\bar{y},\bar{z}) - S_{q\bar{q}g}(x',y,x') - S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) + 1 \tag{3.45}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x',y,x',\bar{x},\bar{y},\bar{z}) = Q(x',x',\bar{z},\bar{x})\,Q(x',y,\bar{y},\bar{z}) = S(\bar{z},\bar{x})\,Q(x',y,\bar{y},\bar{z})$$
$$S_{q\bar{q}g}(x',y,x') = S(x',x')\,S(x',y) = S(x',y)$$
$$S^\dagger_{q\bar{q}g}(\bar{x},\bar{y},\bar{z}) = S(\bar{z},\bar{x})\,S(\bar{y},\bar{z})$$

$$\boxed{\mathcal{W}(x',y,x',\bar{x},\bar{y},\bar{z}) = S(\bar{z},\bar{x})\,Q(x',y,\bar{y},\bar{z}) - S(x',y) - S(\bar{z},\bar{x})\,S(\bar{y},\bar{z}) + 1}$$

- $\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}')$

$$\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y}',\bar{y}') - S_{q\bar{q}g}(x,y,z) - S^\dagger_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') + 1 \tag{3.46}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x,y,z,\bar{x},\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,Q(z,y,\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,S(z,y)$$
$$S_{q\bar{q}g}(x,y,z) = S(x,z)\,S(z,y)$$
$$S^\dagger_{q\bar{q}g}(\bar{x},\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})\,S(\bar{y}',\bar{y}') = S(\bar{y}',\bar{x})$$

$$\boxed{\mathcal{W}(x,y,z,\bar{x},\bar{y}',\bar{y}') = Q(x,z,\bar{y}',\bar{x})\,S(z,y) - S(x,z)\,S(z,y) - S(\bar{y}',\bar{x}) + 1}$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x', y, x') - S^{\dagger}_{q\bar{q}g}(\bar{x}, \bar{y}', \bar{y}') + 1 \tag{3.47}$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = Q(x', x', \bar{y}', \bar{x})\, Q(x', y, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x})\, S(x', y)$$

$$S_{q\bar{q}g}(x', y, x') = S(x', x')\, S(x', y) = S(x', y)$$

$$S^{\dagger}_{q\bar{q}g}(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x})\, S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

$$\boxed{\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x})\, S(x', y) - S(x', y) - S(\bar{y}', \bar{x}) + 1}$$

4 Fourier transform of the dipole terms

New section. At large N_c every Wilson-line structure $\mathcal{W}$ of Sec. 3 is a sum of four kinds of terms: products of two quadrupoles, quadrupole × dipole, terms built only from dipoles, and the constant +1. Here the coordinate integrals are carried out for the *dipole-only* terms, i.e. products of dipoles or a single dipole. The quadrupole terms and the +1 are left untouched. Table 1 lists the terms that are transformed.

channel	Wilson-line structure (multiplies)	dipole-only terms
antiquark	$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ $(+\mathcal{H}[\mathcal{G}; \bar{\mathcal{G}}])$	$-S(x,z)S(z,y)$, $-S(\bar{z},\bar{x})S(\bar{y},\bar{z})$
	$\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$ $(-\mathcal{H}[\mathcal{A}; \bar{\mathcal{G}}])$	$-S(x,y')$, $-S(\bar{z},\bar{x})S(\bar{y},\bar{z})$
	$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$ $(-\mathcal{H}[\mathcal{G}; \bar{\mathcal{A}}])$	$-S(x,z)S(z,y)$, $-S(\bar{y}',\bar{x})$
	$\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$ $(+\mathcal{H}[\mathcal{A}; \bar{\mathcal{A}}])$	$-S(x,y')$, $-S(\bar{y}',\bar{x})$
quark	$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$ $(+\tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \bar{\mathcal{G}}])$	$-S(y,z)S(z,x)$, $-S(\bar{z},\bar{y})S(\bar{x},\bar{z})$
	$\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z})$ $(-\tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \bar{\mathcal{G}}])$	$-S(y,x')$, $-S(\bar{z},\bar{y})S(\bar{x},\bar{z})$
	$\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}')$ $(-\tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \bar{\tilde{\mathcal{A}}}])$	$-S(y,z)S(z,x)$, $-S(\bar{x}',\bar{y})$
	$\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}')$ $(+\tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \bar{\tilde{\mathcal{A}}}])$	$-S(y,x')$, $-S(\bar{x}',\bar{y})$
interference	$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ $(+\mathcal{H}^x[\tilde{\mathcal{G}}; \bar{\mathcal{G}}])$	$-S(x,z)S(z,y)$, $-S(\bar{z},\bar{x})S(\bar{y},\bar{z})$
	$\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z})$ $(-\mathcal{H}^x[\tilde{\mathcal{A}}; \bar{\mathcal{G}}])$	$-S(x',y)$, $-S(\bar{z},\bar{x})S(\bar{y},\bar{z})$
	$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$ $(-\mathcal{H}^x[\tilde{\mathcal{G}}; \bar{\mathcal{A}}])$	$-S(x,z)S(z,y)$, $-S(\bar{y}',\bar{x})$
	$\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}')$ $(+\mathcal{H}^x[\tilde{\mathcal{A}}; \bar{\mathcal{A}}])$	$+S(\bar{y}',\bar{x})S(x',y)$, $-S(x',y)$, $-S(\bar{y}',\bar{x})$

Table 1: The dipole-only terms of the boxed large-N_c structures of Sec. 3. In brackets: the helicity sum the structure multiplies, and its sign in the cross section. The quadrupole terms (QQ, QS) and the constants +1 are not transformed here.

4.1 Assumptions and notation

1. **Homogeneous target.** The dipole depends only on the separation, $S(a,b) = S(a-b)$, as already assumed in the Gaussian expressions $\Gamma(x-z)$ used for the quadrupoles. Its Fourier transform (the momentum-space dipole distribution) is

$$S(a,b) = \int_l e^{i\,l\cdot(a-b)}\,\mathcal{D}(l), \qquad \mathcal{D}(l) = \int d^2r\, e^{-i\,l\cdot r}\, S(r), \qquad \int_l \equiv \int \frac{d^2l}{(2\pi)^2}, \qquad \int_l \mathcal{D}(l) = S(0) = 1. \tag{4.1}$$

Because $\langle S(a,b)\rangle^* = \langle S(b,a)\rangle$, $\mathcal{D}(l)$ is real. For example, the GBW form $S(r) = e^{-Q_s^2 r^2/4}$ gives $\mathcal{D}(l) = \frac{4\pi}{Q_s^2} e^{-l^2/Q_s^2}$. Without an odderon, $\mathcal{D}(l) = \mathcal{D}(-l)$. This is only needed to identify the quark-channel ordering $S(y,z)S(z,x)$ of Sec. 3.2 (which is the complex conjugate of the physical colour structure $S(x,z)S(z,y)$, see Appendix B) with $S(x,z)S(z,y)$. The formulas below are written for the orderings exactly as they appear in Table 1.

2. **Photon transverse momentum.** As in the cross sections of Sec. 3 (which contain no phase $e^{-iq\cdot(c-\bar{c})}$), $q = 0$. For $q \neq 0$, replace $\delta^{(2)}(k_1+k_2+k_3) \to \delta^{(2)}(k_1+k_2+k_3-q)$ below; the relative momenta are unchanged.

3. $S_\perp = \int d^2b$ is the transverse area of the target, and $a \times b \equiv \epsilon^{ij} a^i b^j$.

4.2 Change of variables

Antiquark emission. With $c = \zeta x + \eta y + \xi z$, $R = x - y'$, $Y = y - z$ (Jacobian = 1),

$$x = c + (1-\zeta)R, \qquad y = c - \zeta R + \frac{\xi}{1-\zeta} Y, \qquad z = c - \zeta R - \frac{\eta}{1-\zeta} Y, \qquad y' = c - \zeta R, \tag{4.2}$$

so that the dipole arguments are

$$x - z = R + \frac{\eta}{1-\zeta}\, Y, \qquad z - y = -Y, \qquad x - y' = R, \tag{4.3}$$

and the phase of the cross section separates as

$$k_1\cdot x+k_2\cdot y+k_3\cdot z=K\cdot c+\tilde{k}\cdot R-\tilde{p}\cdot Y,\qquad K=k_1+k_2+k_3,\qquad \tilde{k}=(1-\zeta)k_1-\zeta(k_2+k_3),\qquad \tilde{p}=\frac{\eta\,k_3-\xi\,k_2}{1-\zeta}. \tag{4.4}$$

Quark emission. With c, $\tilde{R}=x'-y$, $X=x-z$, $x'=\frac{\zeta x+\xi z}{\zeta+\xi}$ (Jacobian = 1),

$$y=c-(1-\eta)\tilde{R},\qquad x=c+\eta\tilde{R}+\frac{\xi}{1-\eta}X,\qquad z=c+\eta\tilde{R}-\frac{\zeta}{1-\eta}X,\qquad x'=c+\eta\tilde{R}, \tag{4.5}$$

$$x-z=X,\qquad z-y=\tilde{R}-\frac{\zeta}{1-\eta}\,X,\qquad x'-y=\tilde{R}, \tag{4.6}$$

$$k_1\cdot x+k_2\cdot y+k_3\cdot z=K\cdot c+\tilde{k}_q\cdot\tilde{R}-\tilde{p}_q\cdot X,\qquad \tilde{k}_q=\eta\,(k_1+k_3)-(1-\eta)\,k_2,\qquad \tilde{p}_q=\frac{\zeta\,k_3-\xi\,k_1}{1-\eta}. \tag{4.7}$$

On the support $K=0$ of the results below: $\tilde{k}=k_1$, $\tilde{p}=k_3+\frac{\xi}{1-\zeta}k_1$, $\tilde{k}_q=-k_2$, $\tilde{p}_q=k_3+\frac{\xi}{1-\eta}k_2$. These are the relative momenta of Sec. 2 (photon → parent, parent → daughter + gluon).

4.3 Momentum-space vertex functions

Inverting the Fourier transforms that defined $\mathcal{G},\mathcal{A},\mathcal{G}_E$ in Sec. 2,

$$\mathcal{M}^{lr}_{A\,\lambda_1\lambda_2}(R,Y)=\int\frac{d^2\tilde{k}}{2\pi}\frac{d^2\tilde{p}}{2\pi}\,e^{i\tilde{k}\cdot R-i\tilde{p}\cdot Y}\,\Phi^{lr}_{A\,\lambda_1\lambda_2}(\tilde{k},\tilde{p}),\qquad \int d^2R\,d^2Y\,e^{-i\tilde{k}\cdot R+i\tilde{p}\cdot Y}\,\mathcal{M}_A(R,Y)=(2\pi)^2\,\Phi_A(\tilde{k},\tilde{p}), \tag{4.8}$$

with $A=1$ (before the shock wave, regular + instantaneous) and $A=2$ (after the shock wave):

$$\begin{aligned}
\Phi_1(\tilde{k},\tilde{p})&=\frac{1}{D(\tilde{k},\tilde{p})}\left[\frac{(1-\zeta)\,N(\tilde{k},\tilde{p})}{\tilde{k}^2+\epsilon_f^2}-\xi\,\eta\,I\right],\qquad \Phi_2(\tilde{k},\tilde{p})=\frac{N(\tilde{k},\tilde{p})}{\zeta(1-\zeta)\,(\tilde{k}^2+\epsilon_f^2)\,(\tilde{p}^2+\omega_f^2)},\\
D(\tilde{k},\tilde{p})&=\zeta(1-\zeta)^2(\tilde{p}^2+\omega_f^2)+\xi\,\eta\,(\tilde{k}^2+\epsilon_f^2),\\
N^{lr}_{\lambda_1\lambda_2}(\tilde{k},\tilde{p})&=\sum_\lambda\left[\varphi^{lj}_{\lambda_1\lambda}(\zeta)\tilde{k}^j+m_f\,\mu^l_{\lambda_1\lambda}\right]\left[\tau^{rn}_{\lambda\lambda_2}(\xi,\eta)\tilde{p}^n+\xi\,\omega_f\,\mu^r_{\lambda\lambda_2}\right],\qquad I^{lr}_{\lambda_1\lambda_2}=\left(\delta^{lr}+2i\epsilon^{lr}\lambda_1\right)\delta_{\lambda_1\lambda_2}.
\end{aligned} \tag{4.9}$$

For the quark emission ($\tilde{\mathcal{M}}_A(\tilde{R},X)=\int\frac{d^2\tilde{k}_q}{2\pi}\frac{d^2\tilde{p}_q}{2\pi}e^{i\tilde{k}_q\cdot\tilde{R}-i\tilde{p}_q\cdot X}\,\tilde{\Phi}_A$):

$$\begin{aligned}
\tilde{\Phi}_1(\tilde{k}_q,\tilde{p}_q)&=\frac{1}{\tilde{D}}\left[\frac{(1-\eta)\,\tilde{N}}{\tilde{k}_q^2+\tilde{\epsilon}_f^2}-\xi\,\zeta\,\tilde{I}\right],\qquad \tilde{\Phi}_2(\tilde{k}_q,\tilde{p}_q)=\frac{\tilde{N}}{\eta(1-\eta)\,(\tilde{k}_q^2+\tilde{\epsilon}_f^2)\,(\tilde{p}_q^2+\tilde{\omega}_f^2)},\\
\tilde{D}&=\eta(1-\eta)^2(\tilde{p}_q^2+\tilde{\omega}_f^2)+\xi\,\zeta\,(\tilde{k}_q^2+\tilde{\epsilon}_f^2),\\
\tilde{N}^{lr}_{\lambda_1\lambda_2}&=\sum_\lambda\left[\tau^{nr}_{\lambda_1\lambda}(\xi,\zeta)\tilde{p}_q^n+\xi\,\tilde{\omega}_f\,\mu^r_{\lambda_1\lambda}\right]\left[\varphi^{lj}_{\lambda\lambda_2}(\zeta+\xi)\tilde{k}_q^j+m_f\,\mu^l_{\lambda\lambda_2}\right],\qquad \tilde{I}^{lr}_{\lambda_1\lambda_2}=\left(\delta^{lr}-2i\epsilon^{lr}\lambda_1\right)\delta_{\lambda_1\lambda_2}.
\end{aligned} \tag{4.10}$$

These are, up to the prefactors $-\zeta$ and $+\eta$ of Sec. 2, the momentum-space amplitudes (they agree with the Dirac-spinor amplitude, Appendix B). The combination without any scattering simplifies, because the time of the emission relative to the shock wave becomes irrelevant:

$$\Phi_0\equiv\Phi_1-\Phi_2=-\frac{\xi\,\eta}{D}\left[\frac{N}{\zeta(1-\zeta)(\tilde{p}^2+\omega_f^2)}+I\right],\qquad \tilde{\Phi}_0\equiv\tilde{\Phi}_1-\tilde{\Phi}_2=-\frac{\xi\,\zeta}{\tilde{D}}\left[\frac{\tilde{N}}{\eta(1-\eta)(\tilde{p}_q^2+\tilde{\omega}_f^2)}+\tilde{I}\right]. \tag{4.11}$$

4.4 The basic integral and the dipole kicks

None of the dipole-only terms connects amplitude and conjugate-amplitude coordinates. For such a term, $\mathcal{S}(x,y,z)\,\mathcal{S}'(\bar{x},\bar{y},\bar{z})$, the integrand depends on c and $\bar{c}$ only through the phase. Since $\int d^2c\,d^2\bar{c}\,e^{-iK\cdot(c-\bar{c})}=S_\perp(2\pi)^2\delta^{(2)}(K)$,

$$\begin{aligned}
\int_{x,y,z,\bar{x},\bar{y},\bar{z}}e^{-ik_1\cdot(x-\bar{x})-ik_2\cdot(y-\bar{y})-ik_3\cdot(z-\bar{z})}\,\mathcal{M}_A\,\mathcal{M}_B^*\,\mathcal{S}\,\mathcal{S}'&=S_\perp(2\pi)^2\delta^{(2)}(k_1+k_2+k_3)\,\mathcal{O}_A[\mathcal{S}]\;\left(\mathcal{O}_B[\mathcal{S}'^*]\right)^*,\\
\mathcal{O}_A[\mathcal{S}]\equiv\int d^2R\,d^2Y\,e^{-i\tilde{k}\cdot R+i\tilde{p}\cdot Y}\,\mathcal{M}_A(R,Y)\,\mathcal{S},&\qquad \mathcal{O}_A[1]=(2\pi)^2\Phi_A(\tilde{k},\tilde{p}),
\end{aligned} \tag{4.12}$$

and the same with $(\tilde{R},X,\tilde{k}_q,\tilde{p}_q)$ for the quark channel. On the conjugate side $S(a,b)^*=S(b,a)$, e.g. $[S(\bar{z},\bar{x})S(\bar{y},\bar{z})]^*=S(\bar{x},\bar{z})S(\bar{z},\bar{y})$. A dipole whose argument is $a-b=\alpha R+\beta Y$ multiplies the integrand by $\int_l\mathcal{D}(l)\,e^{il\cdot(\alpha R+\beta Y)}$, i.e. it shifts the momenta of Φ:

$$S(a,b):\quad \Phi(\tilde{k},\tilde{p})\;\to\;\int_l\mathcal{D}(l)\,\Phi(\tilde{k}-\alpha\,l,\;\tilde{p}+\beta\,l). \tag{4.13}$$

From Eqs. (4.3) and (4.6):

antiquark channel	shift	quark channel	shift
$S(x,z)$	$\tilde k \to \tilde k - l,\ \ \tilde p \to \tilde p + \frac{\eta}{1-\zeta} l$	$S(x,z);\ S(z,x)$	$\tilde p_q \to \tilde p_q + l;\ \ \tilde p_q \to \tilde p_q - l$
$S(z,y)$	$\tilde p \to \tilde p - l$	$S(z,y)$	$\tilde k_q \to \tilde k_q - l,\ \ \tilde p_q \to \tilde p_q - \frac{\zeta}{1-\eta} l$
$S(x,y')$	$\tilde k \to \tilde k - l$	$S(y,z)$	$\tilde k_q \to \tilde k_q + l,\ \ \tilde p_q \to \tilde p_q + \frac{\zeta}{1-\eta} l$
		$S(x',y);\ S(y,x')$	$\tilde k_q \to \tilde k_q - l;\ \ \tilde k_q \to \tilde k_q + l$

In terms of the partons, $S(x,z)$ transfers $+l$ to the quark and $-l$ to the gluon, and so on: Φ is evaluated at the momenta *before* the scattering. All the transforms needed are therefore

$$
\begin{aligned}
\Omega_1 &\equiv \mathcal{O}_1[S(x,z)S(z,y)] = (2\pi)^2 \int_{l_1}\int_{l_2} \mathcal{D}(l_1)\mathcal{D}(l_2)\ \Phi_1\Big(\tilde k - l_1,\ \tilde p + \tfrac{\eta}{1-\zeta} l_1 - l_2\Big),\\
\Omega_2 &\equiv \mathcal{O}_2[S(x,y')] = (2\pi)^2 \int_{l} \mathcal{D}(l)\ \Phi_2\Big(\tilde k - l,\ \tilde p\Big),\\
\tilde\Omega_1 &\equiv \tilde{\mathcal{O}}_1[S(y,z)S(z,x)] = (2\pi)^2 \int_{l_1}\int_{l_2} \mathcal{D}(l_1)\mathcal{D}(l_2)\ \tilde\Phi_1\Big(\tilde k_q + l_1,\ \tilde p_q + \tfrac{\zeta}{1-\eta} l_1 - l_2\Big),\\
\tilde\Omega_2 &\equiv \tilde{\mathcal{O}}_2[S(y,x')] = (2\pi)^2 \int_{l} \mathcal{D}(l)\ \tilde\Phi_2\Big(\tilde k_q + l,\ \tilde p_q\Big),\\
\Omega_1^x &\equiv \tilde{\mathcal{O}}_1[S(x,z)S(z,y)] = (2\pi)^2 \int_{l_1}\int_{l_2} \mathcal{D}(l_1)\mathcal{D}(l_2)\ \tilde\Phi_1\Big(\tilde k_q - l_2,\ \tilde p_q + l_1 - \tfrac{\zeta}{1-\eta} l_2\Big),\\
\Omega_2^x &\equiv \tilde{\mathcal{O}}_2[S(x',y)] = (2\pi)^2 \int_{l} \mathcal{D}(l)\ \tilde\Phi_2\Big(\tilde k_q - l,\ \tilde p_q\Big),
\end{aligned}
\tag{4.14}
$$

where in Ω_1 and $\tilde\Omega_1$, l_1 belongs to the first dipole and l_2 to the second. For brevity we write the arguments as

$$
\begin{aligned}
&b = (\tilde k, \tilde p), && a_{l_1 l_2} = \Big(\tilde k - l_1,\, \tilde p + \tfrac{\eta}{1-\zeta} l_1 - l_2\Big), && a_l = (\tilde k - l,\, \tilde p),\\
&\tilde b = (\tilde k_q, \tilde p_q), && \tilde a_{l_1 l_2} = \Big(\tilde k_q + l_1,\, \tilde p_q + \tfrac{\zeta}{1-\eta} l_1 - l_2\Big), && \tilde a_l = (\tilde k_q + l,\, \tilde p_q),\\
& && \hat a_{l_1 l_2} = \Big(\tilde k_q - l_2,\, \tilde p_q + l_1 - \tfrac{\zeta}{1-\eta} l_2\Big), && \hat a_l = (\tilde k_q - l,\, \tilde p_q).
\end{aligned}
\tag{4.15}
$$

(For $\mathcal{D}(l) = \mathcal{D}(-l)$, $\tilde a$ and $\hat a$ give identical integrals, i.e. $\tilde\Omega_A = \Omega_A^x$.)

4.5 Helicity sums in momentum space

For two sets of momenta $a = (k_a, p_a)$ and $b = (k_b, p_b)$ define $\mathfrak{h}_{AB}(a;b) = \sum_{\lambda_1\lambda_2}\sum_{l,r} \Phi^{lr}_{A\,\lambda_1\lambda_2}(a)\,\Phi^{lr\,*}_{B\,\lambda_1\lambda_2}(b)$, and likewise $\tilde{\mathfrak{h}}_{AB}$ (with $\tilde\Phi$, $\tilde\Phi$) and $\mathfrak{h}^x_{AB}$ (with $\tilde\Phi_A(a)$, $\Phi^*_B(b)$; a quark-channel and b antiquark-channel momenta). The building blocks are:
Antiquark emission (N, I):

$$
\begin{aligned}
\mathcal{K}(a;b) \equiv \sum N(a)N^*(b) &= 8\left[\zeta^2 + (1-\zeta)^2\right]\left[\eta^2 + (1-\zeta)^2\right](k_a\cdot k_b)(p_a\cdot p_b) - 8\xi(2\zeta-1)\,(2(1-\zeta)-\xi)\,(k_a\times k_b)(p_a\times p_b)\\
&+ 8m_f^2\left[\eta^2 + (1-\zeta)^2\right](p_a\cdot p_b) + 8\xi^2\omega_f^2\left[\zeta^2 + (1-\zeta)^2\right](k_a\cdot k_b) + 8m_f^2\xi^2\omega_f^2 - 8\zeta\eta\, m_f\xi\omega_f\,(k_a - k_b)\cdot(p_a - p_b),\\
\mathcal{J}(a) \equiv \sum N(a)I^* &= 8\zeta\eta\,(k_a\cdot p_a) - 8m_f\xi\omega_f, \qquad \sum I\,I^* = 8.
\end{aligned}
\tag{4.16}
$$

Quark emission ($\tilde N$, $\tilde I$):

$$
\begin{aligned}
\tilde{\mathcal{K}}(a;b) &= 8\left[\zeta^2 + (1-\eta)^2\right]\left[\eta^2 + (1-\eta)^2\right](k_a\cdot k_b)(p_a\cdot p_b) - 8\xi(2\eta-1)\,(2(1-\eta)-\xi)\,(k_a\times k_b)(p_a\times p_b)\\
&+ 8m_f^2\left[\zeta^2 + (1-\eta)^2\right](p_a\cdot p_b) + 8\xi^2\tilde\omega_f^2\left[\eta^2 + (1-\eta)^2\right](k_a\cdot k_b) + 8m_f^2\xi^2\tilde\omega_f^2 + 8\zeta\eta\, m_f\xi\tilde\omega_f\,(k_a - k_b)\cdot(p_a - p_b),\\
\tilde{\mathcal{J}}(a) &= -8\zeta\eta\,(k_a\cdot p_a) - 8m_f\xi\tilde\omega_f, \qquad \sum \tilde I\,\tilde I^* = 8.
\end{aligned}
\tag{4.17}
$$

Interference ($\tilde N(a)$ with $N^*(b)$):

$$
\begin{aligned}
\mathcal{K}^x(a;b) &= 8\left[1 - \xi - 2\zeta\eta\right]\left[\xi(1-\xi) + 2\zeta\eta\right](k_a\cdot k_b)(p_a\cdot p_b) - 8\xi\left[1 - 2\zeta - \xi\right]^2(k_a\times k_b)(p_a\times p_b)\\
&+ 8m_f^2(\xi + 2\zeta\eta)(p_a\cdot p_b) - 8(\xi + 2\zeta\eta)\,\xi^2\tilde\omega_f\omega_f\,(k_a\cdot k_b) - 8(\zeta+\xi)^2 m_f\xi\omega_f\,(k_a\cdot p_a)\\
&+ 8(1-\zeta)^2 m_f\xi\tilde\omega_f\,(k_b\cdot p_b) - 8\eta^2 m_f\xi\tilde\omega_f\,(k_a\cdot p_b) + 8\zeta^2 m_f\xi\omega_f\,(p_a\cdot k_b),\\
\mathcal{J}^x(a) \equiv \sum \tilde N(a)I^* &= 8(\zeta+\xi)^2(k_a\cdot p_a), \qquad \bar{\mathcal{J}}^x(b) \equiv \sum \tilde I\,N^*(b) = -8(1-\zeta)^2(k_b\cdot p_b), \qquad \sum \tilde I\,I^* = 0.
\end{aligned}
\tag{4.18}
$$

All of these are real. They follow from the coordinate-space sums of Sec. 3 by $R \to -ik_a$, $Y \to ip_a$, $\bar R \to ik_b$, $\bar Y \to -ip_b$ (so the mixed products $R\cdot\bar Y$, $Y\cdot\bar R$ change sign), and were also computed directly. With $A_a = k_a^2 + \epsilon_f^2$, $B_a = p_a^2 + \omega_f^2$, $D_a = \zeta(1-\zeta)^2 B_a + \xi\eta A_a$ and $\tilde A_a = k_a^2 + \tilde\epsilon_f^2$, $\tilde B_a = p_a^2 + \tilde\omega_f^2$, $\tilde D_a = \eta(1-\eta)^2\tilde B_a + \xi\zeta\tilde A_a$:

$$
\begin{aligned}
\mathfrak{h}_{11}(a;b) &= \frac{1}{D_a D_b}\left[\frac{(1-\zeta)^2\mathcal{K}(a;b)}{A_a A_b} - (1-\zeta)\,\xi\eta\left(\frac{\mathcal{J}(a)}{A_a} + \frac{\mathcal{J}(b)}{A_b}\right) + 8\xi^2\eta^2\right], \qquad \mathfrak{h}_{22}(a;b) = \frac{\mathcal{K}(a;b)}{\zeta^2(1-\zeta)^2 A_a B_a A_b B_b},\\
\mathfrak{h}_{12}(a;b) &= \frac{1}{\zeta(1-\zeta)D_a A_b B_b}\left[\frac{(1-\zeta)\mathcal{K}(a;b)}{A_a} - \xi\eta\,\mathcal{J}(b)\right], \qquad \mathfrak{h}_{21}(a;b) = \frac{1}{\zeta(1-\zeta)D_b A_a B_a}\left[\frac{(1-\zeta)\mathcal{K}(a;b)}{A_b} - \xi\eta\,\mathcal{J}(a)\right].
\end{aligned}
\tag{4.19}
$$

$\tilde{\mathfrak{h}}_{AB}$: the same with $\zeta \leftrightarrow \eta$ in the explicit factors ($(1-\zeta) \to (1-\eta)$, $\xi\eta \to \xi\zeta$, $\zeta(1-\zeta) \to \eta(1-\eta)$), $A, B, D \to \tilde{A}, \tilde{B}, \tilde{D}$, $\mathcal{K}, \mathcal{J} \to \tilde{\mathcal{K}}, \tilde{\mathcal{J}}$. Interference:

$$\mathfrak{h}^x_{11}(a;b) = \frac{1}{\tilde{D}_a D_b}\left[\frac{(1-\eta)(1-\zeta)\mathcal{K}^x(a;b)}{\tilde{A}_a A_b} - \frac{(1-\eta)\,\xi\eta\,\mathcal{J}^x(a)}{\tilde{A}_a} - \frac{(1-\zeta)\,\xi\zeta\,\bar{\mathcal{J}}^x(b)}{A_b}\right], \qquad \mathfrak{h}^x_{22}(a;b) = \frac{\mathcal{K}^x(a;b)}{\eta(1-\eta)\zeta(1-\zeta)\tilde{A}_a\tilde{B}_a A_b B_b},$$
$$\mathfrak{h}^x_{12}(a;b) = \frac{1}{\zeta(1-\zeta)\tilde{D}_a A_b B_b}\left[\frac{(1-\eta)\mathcal{K}^x(a;b)}{\tilde{A}_a} - \xi\zeta\,\bar{\mathcal{J}}^x(b)\right], \qquad \mathfrak{h}^x_{21}(a;b) = \frac{1}{\eta(1-\eta)D_b\tilde{A}_a\tilde{B}_a}\left[\frac{(1-\zeta)\mathcal{K}^x(a;b)}{A_b} - \xi\eta\,\mathcal{J}^x(a)\right]. \tag{4.20}$$

Since $\mathcal{K}(a;b) = \mathcal{K}(b;a)$ and everything is real, $\mathfrak{h}_{AB}(b;a) = \mathfrak{h}_{BA}(a;b)$, and the same holds for $\tilde{\mathfrak{h}}$. This symmetry does not hold for $\mathfrak{h}^x$.

4.6 Results

Inserting Eqs. (4.12)–(4.14) term by term (Table 1). Each line gives the dipole term, the $\mathcal{W}$ it comes from, and its contribution to the bracket of the cross section. The common factor $S_\perp(2\pi)^6\delta^{(2)}(k_1+k_2+k_3)$ has been taken out.

Direct, antiquark emission.

$$\begin{aligned}
-S(x,z)S(z,y)\;[\mathcal{W}(x,y,z;\ldots)]:&\;-\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}_{11}(a_{l_1l_2};b), & -S(\bar z,\bar x)S(\bar y,\bar z)\;[\mathcal{W}(x,y,z;\ldots)]:&\;-\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}_{11}(b;a_{l_1l_2}),\\
-S(x,y')\;[\mathcal{W}(x,y',y';\bar x,\bar y,\bar z)]:&\;+\int_l\mathcal{D}\,\mathfrak{h}_{21}(a_l;b), & -S(\bar z,\bar x)S(\bar y,\bar z)\;[\mathcal{W}(x,y',y';\bar x,\bar y,\bar z)]:&\;+\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}_{21}(b;a_{l_1l_2}),\\
-S(x,z)S(z,y)\;[\mathcal{W}(x,y,z;\bar x,\bar y',\bar y')]:&\;+\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}_{12}(a_{l_1l_2};b), & -S(\bar y',\bar x)\;[\mathcal{W}(x,y,z;\bar x,\bar y',\bar y')]:&\;+\int_l\mathcal{D}\,\mathfrak{h}_{12}(b;a_l),\\
-S(x,y')\;[\mathcal{W}(x,y',y';\bar x,\bar y',\bar y')]:&\;-\int_l\mathcal{D}\,\mathfrak{h}_{22}(a_l;b), & -S(\bar y',\bar x)\;[\mathcal{W}(x,y',y';\bar x,\bar y',\bar y')]:&\;-\int_l\mathcal{D}\,\mathfrak{h}_{22}(b;a_l).
\end{aligned} \tag{4.21}$$

Here $\int_{l_1l_2}\mathcal{D}\mathcal{D} \equiv \int_{l_1}\int_{l_2}\mathcal{D}(l_1)\mathcal{D}(l_2)$. The right column is the left column with amplitude and conjugate amplitude exchanged; with $\mathfrak{h}_{AB}(b;a) = \mathfrak{h}_{BA}(a;b)$ the sum of all eight terms is

$$\begin{aligned}
\left.\frac{d^3N^{\rm direct}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\rm dipole} &= \frac{e^2\,g^2\,\zeta^2\,C_F\,N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+\,2}}\,\delta\left(q^+-k_1^+-k_2^+-k_3^+\right)\delta^{(2)}(k_1+k_2+k_3)\sum_f e_f^2\\
&\quad\times(-2)\left\{\int_{l_1}\int_{l_2}\mathcal{D}(l_1)\mathcal{D}(l_2)\Big[\mathfrak{h}_{11}-\mathfrak{h}_{12}\Big](a_{l_1l_2};b) - \int_l\mathcal{D}(l)\Big[\mathfrak{h}_{21}-\mathfrak{h}_{22}\Big](a_l;b)\right\}\\
&= \ldots\times(-2)\sum_{\lambda_1\lambda_2lr}\frac{(\Omega_1-\Omega_2)\,\Phi_0^*(b)}{(2\pi)^2}.
\end{aligned} \tag{4.22}$$

Direct, quark emission. The same with tildes:

$$\begin{aligned}
\left.\frac{d^3N^{\rm direct}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\rm dipole} &= \frac{e^2\,g^2\,\eta^2\,C_F\,N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+\,2}}\,\delta\left(q^+-k_1^+-k_2^+-k_3^+\right)\delta^{(2)}(k_1+k_2+k_3)\sum_f e_f^2\\
&\quad\times(-2)\left\{\int_{l_1}\int_{l_2}\mathcal{D}(l_1)\mathcal{D}(l_2)\Big[\tilde{\mathfrak{h}}_{11}-\tilde{\mathfrak{h}}_{12}\Big](\tilde a_{l_1l_2};\tilde b) - \int_l\mathcal{D}(l)\Big[\tilde{\mathfrak{h}}_{21}-\tilde{\mathfrak{h}}_{22}\Big](\tilde a_l;\tilde b)\right\},
\end{aligned} \tag{4.23}$$

term by term exactly as for the antiquark ($a \to \tilde a$, $b \to \tilde b$, $\mathfrak{h} \to \tilde{\mathfrak{h}}$; here l_1 belongs to $S(y,z)$ and l_2 to $S(z,x)$).

Interference.

$$\begin{aligned}
-S(x,z)S(z,y)\;[\mathcal{W}(x,y,z;\bar x,\bar y,\bar z)]:&\;-\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}^x_{11}(\hat a_{l_1l_2};b),\\
-S(\bar z,\bar x)S(\bar y,\bar z)\;[\mathcal{W}(x,y,z;\bar x,\bar y,\bar z)]:&\;-\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}^x_{11}(\tilde b;a_{l_1l_2}),\\
-S(x',y)\;[\mathcal{W}(x',y,x';\bar x,\bar y,\bar z)]:&\;+\int_l\mathcal{D}\,\mathfrak{h}^x_{21}(\hat a_l;b),\\
-S(\bar z,\bar x)S(\bar y,\bar z)\;[\mathcal{W}(x',y,x';\bar x,\bar y,\bar z)]:&\;+\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}^x_{21}(\tilde b;a_{l_1l_2}),\\
-S(x,z)S(z,y)\;[\mathcal{W}(x,y,z;\bar x,\bar y',\bar y')]:&\;+\int_{l_1l_2}\mathcal{D}\mathcal{D}\,\mathfrak{h}^x_{12}(\hat a_{l_1l_2};b),\\
-S(\bar y',\bar x)\;[\mathcal{W}(x,y,z;\bar x,\bar y',\bar y')]:&\;+\int_l\mathcal{D}\,\mathfrak{h}^x_{12}(\tilde b;a_l),\\
-S(x',y)\;[\mathcal{W}(x',y,x';\bar x,\bar y',\bar y')]:&\;-\int_l\mathcal{D}\,\mathfrak{h}^x_{22}(\hat a_l;b),\\
-S(\bar y',\bar x)\;[\mathcal{W}(x',y,x';\bar x,\bar y',\bar y')]:&\;-\int_l\mathcal{D}\,\mathfrak{h}^x_{22}(\tilde b;a_l),\\
+S(\bar y',\bar x)S(x',y)\;[\mathcal{W}(x',y,x';\bar x,\bar y',\bar y')]:&\;+\int_l\int_{l'}\mathcal{D}(l)\mathcal{D}(l')\,\mathfrak{h}^x_{22}(\hat a_l;a_{l'}),
\end{aligned} \tag{4.24}$$

where l_1 belongs to $S(x,z)$ and l_2 to $S(z,y)$ (on the conjugate side, to $S(\bar z,\bar x)$ and $S(\bar y,\bar z)$). The last term is the only dipole-only term with Wilson lines on both sides. It arises because the two after-shock-wave emissions come from different partons, so the colour trace splits into two dipoles; in the direct terms the corresponding structure is a quadrupole. In total:

$$\begin{aligned}\left.\frac{d^3N^{\rm cross}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\rm dipole} &= -\frac{e^2\,g^2\,\eta\,\zeta\,C_F\,N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+\,2}}\,\delta\left(q^+-k_1^+-k_2^+-k_3^+\right)\delta^{(2)}(k_1+k_2+k_3)\sum_f e_f^2\\
&\times 2\,\Re\Bigg\{-\int_{l_1}\int_{l_2}\mathcal{D}(l_1)\mathcal{D}(l_2)\Big[\mathfrak{h}^x_{11}-\mathfrak{h}^x_{12}\Big](\hat a_{l_1l_2};b)\\
&+\int_l\mathcal{D}(l)\Big[\mathfrak{h}^x_{21}-\mathfrak{h}^x_{22}\Big](\hat a_l;b)-\int_{l_1}\int_{l_2}\mathcal{D}(l_1)\mathcal{D}(l_2)\Big[\mathfrak{h}^x_{11}-\mathfrak{h}^x_{21}\Big](\tilde b;a_{l_1l_2})+\int_l\mathcal{D}(l)\Big[\mathfrak{h}^x_{12}-\mathfrak{h}^x_{22}\Big](\tilde b;a_l)\\
&+\int_l\int_{l'}\mathcal{D}(l)\mathcal{D}(l')\,\mathfrak{h}^x_{22}(\hat a_l;a_{l'})\Bigg\}.\end{aligned}\tag{4.25}$$

In compact form the bracket is $-(\Omega_1^x-\Omega_2^x)(2\pi)^2\Phi_0^*(b)-(2\pi)^2\tilde\Phi_0(\tilde b)(\Omega_1-\Omega_2)^*+\Omega_2^x\,\Omega_2^*$, summed over helicities and polarisations and divided by $(2\pi)^4$.

4.7 Checks

1. **No target** ($S\to1$, i.e. $\mathcal{D}(l)\to(2\pi)^2\delta^{(2)}(l)$): $\Omega_A\to(2\pi)^2\Phi_A(b)$, and the dipole part of the antiquark direct term becomes $-2\sum|\Phi_0(b)|^2$. In each $\mathcal{W}$ the dipole-only terms then add up to -2 (interference $\mathcal{W}(x',y,x';\bar x,\bar y',\bar y')$: $+1-1-1=-1$). The quadrupole terms and the $+1$ add up to $+2$ (resp. $+1$), so the cross section vanishes without a target, as it must.
2. **Massless limit:** $m_f\to0$ removes all helicity-flip terms of $\mathcal{K}$, $\tilde{\mathcal{K}}$, $\mathcal{K}^x$, $\mathcal{J}$; with $\epsilon_f^2\to\zeta(1-\zeta)Q^2$ and $\omega_f\to0$ the massless results follow.
3. The shift rules were derived symbolically and tested numerically with Gaussian test functions (agreement 10^{-11}), and all $\mathfrak{h}_{AB}$, $\tilde{\mathfrak{h}}_{AB}$, $\mathfrak{h}^x_{AB}$ were compared with brute-force helicity sums of Φ, $\tilde\Phi$ (agreement 10^{-15}), Appendix B.

A Corrections to the supplied file

For each item: [**equation number(s) and page(s) in this PDF**], then the line number(s) in the supplied `Trijet_massive.tex`. In the marked version the changes are printed in red. Apart from these items, the file is identical to the corrected version checked previously, and all of its results were re-verified (Appendix B).

Errors

1. [**Eq.** (2.20)**, p. 5**] (L224): in the after-shock-wave transform the exponent must be $e^{-i\tilde p\cdot Y}$ (the phase of the state, L143, and the $\tilde p$ transform of L221), not $e^{+i\tilde p\cdot Y}$. With $e^{+i\tilde p\cdot Y}$ the right-hand side would carry $+iY^n$ instead of $-iY^n$ in the $\mathcal{A}^{(01)}$ term. This had been fixed before and was reintroduced.
2. [**Eq.** (3.30)**, p. 12**] (L550): in the double-flip term of the quark-channel after-shock-wave vertex $\tilde\Psi[\tilde A]$ the gluon-flip factor is $\xi\tilde\omega_f=\xi^2m_f/(1-\eta)$, not $\xi\omega_f=\xi^2m_f/(1-\zeta)$. It must match $\tilde\Psi[\tilde{\mathcal{G}}]$, the second term of the same line, and the Dirac-spinor result for the quark line.
3. [**Eq.** (2.19)**, p. 4; Eq.** (2.39)**, p. 7**] (L213, L344): energy denominator $-E_q(k'')+E_{\bar q}(p'')\to-E_q(k'')-E_{\bar q}(p'')$ (reintroduced). The reduced denominators used afterwards were already correct.
4. [**Eq.** (3.23)**, p. 12**] (L522): $\tilde{\mathcal{M}}_1(\tilde{\mathcal{G}})$ was written twice in a row, which reads as a product $\tilde{\mathcal{M}}_1\tilde{\mathcal{M}}_1$.

Wording and notation

5. [**p. 5**] (L241–L242): "$(b)=(a)|_{y,z\to y'}$" holds literally only for $m_f=0$. With masses $\mathcal{A}^{(\alpha\beta)}$ are different functions from $\mathcal{G}^{(\alpha\beta)}$; only their $Y\to0$ limits agree.
6. [**p. 6**] (L265–L266): the sentence refers to the factor $\frac{Y^2}{\bar R\cdot Y}$ of the massless effective vertex, which no longer appears; rephrased.
7. [**p. 6**] (L297): "$D\to\tilde Q\tilde R$" is the massless after-shock-wave rule, and for the quark channel the photon scale is $\eta(1-\eta)Q^2$, not $\tilde Q^2=\zeta(1-\zeta)Q^2$.
8. [**Eq.** (2.40)**, p. 8**] [**Eq.** (3.24)**, p. 12**] [**p. 12**] (L348, L525, L532): $[\tilde A]\to[\bar A]$, $(\tilde A)\to[\bar A]$, and the argument of $\tilde{\mathcal{G}}^{(\alpha\beta)}$ is $(\tilde R,X)$, not (R,X).
9. [**p. 12; Eq.** (3.39)**, p. 14; Eq.** (3.40)**, p. 14**] (L535, L613, L617): "Explicilty"; "," → ";" in $\mathcal{H}^x[\cdot\,;\cdot]$.

Not an error, but an assumption to state: the quark-direct structures are written as $\mathcal{W}(y,x,z,\bar y,\bar x,\bar z)$ etc. They are the complex conjugates of the physical colour structures of the quark-emission amplitudes (Appendix B, check J), and equal to them only for a C-even target, $\langle\mathcal{W}\rangle$ real. This is the case for the Gaussian (MV-type) averages used here. It is also why the quark-direct dipole terms appear as $S(y,z)S(z,x)$ rather than $S(x,z)S(z,y)$ in Sec. 4.

Overall normalisation: the prefactors $\frac{e^2g^2\zeta^2C_FN_c}{8(2\pi)^{12}\xi q^{+2}}$ etc. are those of the massless note; the mass does not change them. The checks verify everything relative to them (signs and weights of all terms, channels and helicity paths), not the absolute normalisation itself.

B How the results were checked

The scripts are in `trijet/checks/massive/` of the repository.

E–I (unchanged parts of Secs. 1–3, from the previous round) Massive two-component vertices fitted to Lepage–Brodsky spinors; the complete amplitude (both emitters, regular + instantaneous) equal to the Feynman/Dirac-spinor amplitude for all helicities and polarisations; the radial functions $\mathcal{G}^{(\alpha\beta)}$, $\mathcal{A}^{(\alpha\beta)}$, $\mathcal{G}_E$ against an independent Hankel evaluation (10^{-13}); all helicity sums $\mathcal{H}$, $\tilde{\mathcal{H}}$, $\mathcal{H}^x$ computed symbolically.

J `checkJ_colour.py`: every large-N_c Wilson-line structure of Sec. 3 against the exact colour trace $\frac{1}{C_F N_c}\sum_a \mathrm{Tr}[C_A^a C_B^{a\dagger}]$ of the amplitude colour matrices ($U^{ab}(z)V(x)t^bV^\dagger(y)-t^a$, $V(x)V^\dagger(y')t^a-t^a$, $t^aV(x')V^\dagger(y)-t^a$) for random $SU(N)$ matrices, $N=4,\dots,32$. The difference falls exactly like $1/N^2$ for all twelve structures (the two pure after-shock-wave direct structures are exact for every N). The quark-direct structures as written equal the complex conjugates of the physical ones.

K `checkK_dipole_ft.py`: the changes of variables (Jacobian = 1), the phase decompositions (4.4), (4.7), the dipole arguments (4.3), (4.6) and the shift table, symbolically; the identity $\Phi_1-\Phi_2=\Phi_0$, Eq. (4.11); and a numerical test of the convolution formula for Ω_1 with Gaussian test functions, coordinate integral vs momentum convolution, agreement 2×10^{-11}.

L `checkL_momentum_sums.py`: $\mathcal{K},\mathcal{J},\tilde{\mathcal{K}},\tilde{\mathcal{J}},\mathcal{K}^x,\mathcal{J}^x,\bar{\mathcal{J}}^x$ of Eqs. (4.16)–(4.18), symbolically in ζ,ξ,m_f and for independent momenta $a\neq b$.

M `checkM_h_closed_forms.py`: all twelve closed forms $\mathfrak{h}_{AB}$, $\tilde{\mathfrak{h}}_{AB}$, $\mathfrak{h}^x_{AB}$ of Eqs. (4.19), (4.20) against brute-force sums of the vertex functions (4.9), (4.10) over λ_1,λ_2,l,r, for random momenta, masses (including $m_f=0$) and fractions: agreement 10^{-15}.